

Intracardiac ECG Pulse Localization using Overlapping Block Sparse Reconstruction

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Abstract

The processing and analysis of multichannel signals which exhibit some synchronicity is of critical importance in biomedical signal processing. Though sparse reconstruction of such signals is well known, penalties enforcing both sparsity and partial synchronicity between channels have been less investigated. In this paper, we present an algorithm based on overlapping grouped LASSO to guarantee both sparsity and synchronicity for multichannel signals. We show that the proposed method can be implemented efficiently, using consensus learning approaches and recent work of the authors for accelerating the related optimization procedure. Results on synthetic data show the efficiency of the proposed approach. When applied on real intracardiac recordings, the proposed method succeeds in automatically detecting electrical pulses both for sinus rhythm and atrial fibrillation episodes.

Keywords: Sparse reconstruction; Grouped LASSO; ECG signal processing; multichannel signal processing

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1. Introduction

Electrocardiographic (ECG) signals were initially recorded in humans by Augustus D. Waller in 1887. Soon afterwards, Willem Einthoven substantially improved the recording mechanism, named the main deflections in the ECG (the now presently named P and T waveforms, and the QRS complex), and showed that several cardiovascular disorders were related to specific features in the ECG [1, 2]. Since then, multi-channel ECGs have become the main biomedical signals used to diagnose cardiac pathologies [3, 4, 5], whereas single-channel ECGs are also routinely acquired and exploited in other non-cardiac, health-related contexts, like clinical EEG recordings [6, 7] or the monitoring of patients with sleep disorders [8]. This widespread use of ECG recordings heavily relies on the development of signal processing and machine learning tools for noise removal and interference suppression [9, 10], feature extraction [11, 12], and computer-aided diagnosis of diseases [13, 14, 15].

During the last two decades, the use of intracardiac or intracavitary ECGs (a.k.a. electrograms (EGMs)), recorded by placing electrodes inside the heart in direct contact with the myocardium’s wall, has become more and more relevant. For instance, EGM-guided RF catheter ablation of atrial fibrillation (AF) is now considered a safe and effective therapeutic alternative to the pharmacological approach [16, 17]. Catheter ablation was initially used only for patients with chronic AF that could not be treated with drug therapy, but currently it is more and more commonly used as the chosen therapy (either as first or second-line option) for other types of AFs [18, 19, 20]. Moreover, modern cardiac implantable devices (i.e., pacemakers and cardioverter defibrillators) are able to store and transmit EGMs that can later be visualized by cardiologists to verify device function, monitor patient condition and validate diagnostics [21, 22, 23].

In all cases, it is essential to develop fast and efficient (both in terms of memory storage and energy consumption) algorithms for multi-channel ECG/EGM signal processing. On the one hand, implantable devices have very limited storage and transmission capabilities, as well as strict power consumption re-

31 requirements [24], so it is crucial to perform an efficient real-time detection of the
 32 relevant heartbeats (i.e., the ones related to a pathological cardiac condition)
 33 to mark them for storage and/or transmission. On the other hand, EGMs are
 34 used in the electrophysiology laboratory to detect candidate locations for RF
 35 ablation (e.g., sites with high dominant frequencies or fractionated EGMs), so
 36 real-time or quasi-real-time extraction of the relevant features from the train of
 37 activations (i.e., heartbeats) is also required. Finally, although real-time pro-
 38 cessing may also be required in the case of external ECGs (e.g., in intensive
 39 care units (ICUs) [25]), the signal processing of long ECG recordings will also
 40 benefit from fast and efficient algorithms, even if no real-time requirements ex-
 41 ist, by increasing the size of the segments that can be processed simultaneously,
 42 reducing computation times and decreasing the amount of memory needed.

43 In recent contributions, the localization of peaks in single channel ECGs and
 44 EGMs was performed using sparse reconstruction techniques, either by using a
 45 thresholded LASSO [26, 27] or by incorporating further restrictions into the
 46 LASSO formulation [28, 29]. Though interesting in practice for single channel
 47 signals, the main limitation of these methods was that they did not take into
 48 consideration the synchronicity between channels for multi-channel signals. In
 49 order to illustrate this situation, Figure 1 shows two examples of multi-channel
 50 EGMs. Note the existing differences between one intracardiac recording sinus
 51 rhythm (left, blue curves), and a AF episode (right, red curves). Among others,
 52 the main differences are longer inter-pulse intervals and higher pulse amplitudes
 53 in the normal case. Additionally, electrical pulses in the AF case are more
 54 widespread. Furthermore, in both cases additional artifacts stemming from
 55 the movement of the electrodes may appear (e.g., see the second and third
 56 blue curves), making the automatic pulse localization task more challenging.
 57 However, in both cases it is clear that pulse localization techniques could greatly
 58 benefit from the additional knowledge about channels' synchronicity.

59 A first attempt to include this synchronism into the reconstruction is detailed
 60 in [30]. It uses Grouped LASSO (GL) instead of LASSO, with groups based
 61 on identical sampling points from all the channels. However, this approach

62 yields suboptimal results, mainly because grouping single times altogether is
 63 too drastic and leads to numerous wrong detections when some electrode motion
 64 noise is wrongly detected in one single channel. Note that the use of the grouped
 65 LASSO for propagation pattern analysis from unipolar and bipolar EGMs was
 66 already proposed in [31, 32, 33]. However, they applied the GL to obtain sparse
 67 coefficients in the MVAR (Multivariate Autoregressive) model which is fitted to
 68 the multi-channel EGMs, rather than directly on the signals to detect the sparse
 69 collection of peaks associated to the wavefront. The MVAR approach can lead
 70 to more interpretable propagation maps, but also imposes some assumptions on
 71 the wave propagation which may not be true in some cases and which do not
 72 appear when working directly with the EGMs. The problem of synchronous
 73 information extraction from multivariate data also appears in numerous fields,
 74 from acoustics [34] to general multivariate time series analysis [35]. The use of
 75 machine learning and artificial neural networks is nowadays common to extract
 76 this information. For example, [35] used a two step neural network approach
 77 (convolutional layers to extract the spatial correlations, and LSTM (Long Short-
 78 Term Memory) layers to catch the temporal correlations) to perform inference
 79 on multivariate time series. Though these tools are appealing for numerous
 80 applications, they cannot be applied in a straightforward manner for intra-
 81 cardiac ECG recordings, mostly due to the lack of enough data to train the
 82 networks.

83 In this paper, we propose to relax the constraint on the grouped LASSO
 84 described above, by building groups of close, consecutive sampling times from
 85 all the channels. To maximize the precision of the localisation, the groups may
 86 overlap, and thus the optimization problem, though similar, cannot be solved
 87 using the same techniques. Several approaches have been suggested to find the
 88 overlapping grouped LASSO (OGL) minimizer, as described in [36] and refer-
 89 ences therein. However, it has been recently shown (as recalled in [37]) that
 90 the OGL solution can be computed using a consensus-based learning approach,
 91 by means of the Alternating Direction Method of Multipliers (ADMM). In this
 92 paper, we derive the ADMM solution for OGL, and extend the acceleration

method for ADMM detailed in [38, 39] so that it can be used for OGL to efficiently compute this solution. Therefore, the novel contributions of this paper are twofold. From a methodological standpoint, we introduce an effective, novel OGL formulation for ECG pulse localization in multichannel intra-cardiac ECG signals. We also show, both on synthetic and real data, that the introduction of overlapping in grouped variables allows to significantly improve the pulse localization and reconstruction. From a practical standpoint, we extend the approach presented in [39], such that the method developed in this contribution can be implemented for near real-time applications related to biomedical engineering.

Note that our motivation is to provide an efficient and robust method to detect heartbeats from EGM recordings by exploiting the available multi-channel information. Although we do not focus on any particular application, several problems in cardiology where EGMs are used could benefit from the proposed methodology. For example, the suboptimal outcomes of RF ablation of AF patients urges the development of advanced signal processing techniques to construct interpretable propagation maps that can help cardiologists to better determine candidate areas for ablation [40]. Although many different approaches to construct these propagation maps have been proposed during the last 5 years [41, 42], there is still no widely accepted methodology. In this sense, by providing a robust method to automatically detect the peaks that define the wavefront, our approach could serve as the basis for a more reliable method to construct those propagation maps. As a second example, let us consider implantable devices like pacemakers or cardioverter defibrillators (ICDs). These devices have substantially evolved during the last decade, including enhanced data storage and transmission capabilities, as well as several diagnostic features [43, 44, 45]. However, ICDs still record data (from the 2–4 available EGM channels) either periodically at pre-scheduled time intervals or on patient demand, rather than on an event-triggered basis. A robust method that can automatically detect relevant activations in the multi-channel EGM signal and store them for further revision by the cardiologist would be a very relevant addition to current ICDs.

124 The rest of the paper is organized as follows. Section 2 describes the model
125 used, and how it can be related to the OGL problem. Section 3 details how
126 the OGL minimizer can be found by using the acceleration of [39], and the
127 post-processing used to estimate the electrical pulses' localization. Results on
128 synthetic data - to investigate the influence of the hyperparameters used - and
129 on real data are presented in Section 4. [For the sake of clarity, Table 1 provides
a list of the acronyms used throughout the text.](#)

Acronym	Meaning
ADMM	Alternated Direction Method of Multipliers
AF	Atrial Fibrillation
ECG	ElectroCardioGram
EEG	ElectroEncephaloGram
EGM	ElectroGraM (a.k.a. intracardiac/intra-cavitary ECG)
FFT	Fast Fourier Transform
GL	Group LASSO
ICD	Implantable Cardioverter Defibrillator
ICU	Intensive Care Unit
LASSO	Least Absolute Shrinkage and Selection Operator
LSTM	Long Short-Term Memory
MVAR	MultiVariate AutoRegressive
OGL	Overlapping Group LASSO
RF	RadioFrequency
SVD	Singular Value Decomposition

Table 1: List of acronyms used in the text.

130

131 2. Model Overview

132 In this section, we present the mathematical model used in this paper, and
133 we illustrate the existing analogy between the problem at hand and the grouped
134 sparse regression framework.

135 2.1. Model Description

136 As stated above, we observe synchronized channels stemming from several
 137 electrodes. Namely, we observe N samples of M digitized signals defined as
 138 follows:

$$y_i[n] = \sum_{k=-\infty}^{\infty} a_{k,i} \Phi_{k,i}[n - m_{k,i}] + \varepsilon_i[n], \quad n = 0 \dots N - 1, \quad i = 1 \dots M \quad (1)$$

139 where $\Phi_{k,i}$ denotes the k -th electrical pulse observed in the i -th channel, $a_{k,i}$
 140 its amplitude, $m_{k,i}$ the sampled time from which the k -th pulse starts, and
 141 ε_i is additive white Gaussian noise. We assume that the baseline and other
 142 existing interferences have been removed from the digital signal, e.g. by means
 143 of standard digital signal processing techniques, as detailed in [46]. Furthermore,
 144 due to the nature of the investigated signals, we also assume that the shapes $\Phi_{k,i}$
 145 have a compact support $L \ll N$, so that the sum in (1) can be transformed into a
 146 finite sum. This implies that only a finite number of shapes are actually present
 147 or partly present in all the signals $y_1, y_2, \dots, y_M$; we further on denote by p
 148 this number. The aim of this paper is to present a method for the estimation of
 149 the sampled times, m_k , and, to a lesser extent, of the pulses' amplitudes $a_{k,i}$.

150 Following is a definition which will be used throughout the paper:

151 **Definition 1 (δ -synchronicity).** *Given $\delta > 0$, we say that the signals $y_1, y_2,$
 152 $\dots, y_M$ are δ -synchronous if for all integer k , there exists an interval $V_k(\delta)$ of
 153 length δ such that $m_{k,i} \in V_k(\delta)$ for all $i = 1 \dots M$. Intervals $V_k(\delta)$ are called
 154 δ -neighborhoods.*

155 δ -synchronicity guarantees that electrical pulses occur more or less simulta-
 156 neously, enforcing that all the pulses associated with the same electrical activity
 157 are at a distance of at most δ samples from each other. Throughout the paper
 158 we will assume that the multichannel signal described in (1) is δ -synchronous.
 159 This assumption can be physiologically justified, since catheters used for the
 160 recording of intracardiac ECGs measure electrical activity in very local areas
 161 of the heart. Note that a preliminary estimation of δ , based on material con-
 162 straints (shape of the catheter, conduction velocity and sampling frequency of

the device), is required in practice. However, such an estimate of the maximum value of δ can be easily obtained as

$$\delta = \left\lceil \frac{T_d^{\max}}{T_s} \right\rceil = \left\lceil \frac{d_{\max}}{v_{\min} T_s} \right\rceil,$$

where $\lceil x \rceil$ denotes the integer part of x rounded up, $T_d^{\max} = \frac{d_{\max}}{v_{\min}}$ is the maximum delay among any two channels, $d_{\max}$ is the maximum distance among two electrodes, $v_{\min}$ is the minimum conduction velocity inside the heart, and T_s is the sampling period.

2.2. Overlapping Grouped LASSO for Grouped Sparsity

Should we be interested in the reconstruction of one single channel and the localisation of single electrical activations in it, the problem could then be considered as a sparse regression problem. The sparse aspect in the latter can be justified by the existence of a physiological refractory period between two electrical pulses. In this framework, it has been shown (e.g., in [26]) that pulses in individual channels could be efficiently localised in the case of sinus rhythm using a thresholded LASSO. However, this approach has several shortcomings when dealing with multichannel signals. Among other things, it does not take into account the inherent correlation between channels. More specifically, the electrodes recording the electrical signals on each channel are very close spatially; therefore, electrical pulses on different channels start at very close sampling points one from each other.

In a previous contribution [30], the authors suggested using a grouped sparsity paradigm, by creating groups of sampling points to enforce grouped activations of the arrival times. However, in light of the results detailed in [30], this approach is much too coarse. The following paragraphs recall and extend this paradigm to make it more usable and attain better performance. The underlying rationale for the proposed model extension is that we can compromise on the accuracy in the estimation of the arrival times, provided the sampling frequency is sufficiently high, in order to attain an increased robustness. This allows us to group consecutive sampling points inside the same group of variables. This

191 can be seen as an artificial downsampling of the data at hand, when it comes
 192 to estimation of arrival times.

193 We now detail the model extension mathematically. Let us denote by

$$\mathbf{y} = [y_1[0] \ y_1[1] \ \dots y_1[N-1] \ y_2[0] \ \dots y_2[N-1] \ \dots y_M[0] \ \dots y_M[N-1]]^T$$

194 the concatenation of all the channels into a single $NM \times 1$ vector; and by

$$\boldsymbol{\varepsilon} = [\varepsilon_1[0] \ \varepsilon_1[1] \ \dots \varepsilon_1[N-1] \ \varepsilon_2[0] \ \dots \varepsilon_2[N-1] \ \dots \ \varepsilon_M[0] \ \dots \ \varepsilon_M[N-1]]^T$$

195 the vectorization of the noise contribution, based on the same argument. Re-
 196 calling that p is the total number of pulse shapes present in $\mathbf{y}$, we denote by $\mathbf{B}$
 197 the $MN \times (MN - L)p$ matrix which defines the dictionary of template shapes
 198 with common support size L used to characterize $\mathbf{y}$. Assuming that the signals
 199 can be expressed as a linear combination of the columns of $\mathbf{B}$, the model we
 200 eventually investigate is

$$\mathbf{y} = \mathbf{B}\mathbf{x} + \boldsymbol{\varepsilon},$$

201 where $\mathbf{x}$ is a column vector of length $(MN - L)p$, which can be used to infer
 202 both location and intensity of the pulses.

203 If the p pulse shapes appearing in the signal are perfectly known, the vector
 204 $\mathbf{x}$ is naturally sparse, and its non-zero coefficients indicate the sampling points
 205 from which the pulses begin. However, in practice the shapes actually appearing
 206 in the signal are unknown. Finding a dictionary which is well suited for the
 207 data is a challenging issue in all compressive sensing and sparse approximation
 208 applications. Several suggestions have been done in the past years in order
 209 to estimate ECG signals in a satisfactory manner. For example, [26, 30] used
 210 a dictionary made of translated Gaussian waveforms, whereas [27] suggested
 211 using a dictionary composed of Mexican hat waveforms. Automatic dictionary
 212 learning has also been suggested by means of the K-SVD algorithm [47], or
 213 more recently by several approaches that learn the ECG shapes directly from the
 214 ECG signals [48, 49, 50, 51]. Although such approaches can yield good results
 215 on extra-cardiac ECG signals, their use on multichannel intra-cardiac signals is
 216 more problematic, due to the large variability in the shape of the activations

217 between channels and even between pulses. Therefore, in this paper, we have
 218 used a generic dictionary of p truncated and translated Gaussian waveforms,
 219 with a common support of L sampling points, spanning over all the channels
 220 altogether. For one single shape k , we build the sub-dictionary $\mathbf{A}_k$, defined as

$$\mathbf{A}_k = \left[\exp \left(-\frac{(n-m)^2}{2\sigma_k^2} \right) \right]_{1 \leq n \leq NM, 1 \leq m \leq NM-L}, \quad k = 1 \dots p.$$

221 Then we built the global dictionary $\mathbf{B}$ as the block diagonal matrix obtained by
 222 replicating $\mathbf{A}$:

$$\mathbf{B} = [\mathbf{A}_1 \ \mathbf{A}_2 \ \dots \ \mathbf{A}_p]$$

223 As previously detailed, our objective is to obtain a regressor $\hat{\mathbf{x}}$, whose sparsity
 224 ensures δ -synchronism between channels. Group LASSO (GL) provides a sat-
 225 isfactory answer for this objective, since it was introduced as a refinement of
 226 LASSO in order to enforce grouped sparsity [52]. The GL regressor is defined
 227 as

$$\hat{\mathbf{x}}_{GL}(r) = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathbf{B}\mathbf{x}\|_2^2 + r \sum_{G \in \mathcal{G}} \|\mathbf{x}\|_{2,G} \quad (2)$$

228 where r in (2) is a sparsity parameter which controls the trade-off between pre-
 229 cision of the reconstruction and grouped sparsity, $\mathcal{G}$ is the set of groups (i.e., a
 230 set of integer intervals $G = \{i_1, i_2, \dots, i_g\} \subset \{0, \dots, N-1\}$ whose union equals
 231 $\{0, \dots, N-1\}$), and $\|\mathbf{x}\|_{2,G}^2 = \sum_{n \in G} x_n^2$. Note that [52] defined the groups as a
 232 partition (thus, each variable would appear only once in the reconstruction),
 233 whereas we allow the groups to overlap. Although the proposed approach is
 234 generic, and the group definitions can be made arbitrarily, in this paper we
 235 define each group as the set of consecutive sampling points, related to every
 236 channel with a certain amount of overlap with respect to the previous and pos-
 237 terior groups. This choice allows us to relate the groups with the δ -synchronicity
 238 previously defined, as later shown in Section 4 In our framework, this overlap-
 239 ping between groups introduces an additional difficulty in the optimization of
 240 the cost function. Indeed, an out-of-the-box application of Group LASSO in
 241 that matter would yield misinterpretations in the reconstruction, and error in

the peaks' localization. It is therefore required to take the overlapping of the created groups into account when solving the optimization problem in (2).

3. Algorithm for Fast OGL Resolution and Pulse Localization

Before introducing the details of the proposed approach, we summarize it in the next section.

3.1. Algorithm Overview

The algorithmic pipeline can be summarized with the block diagram presented in Figure 2. The cornerstone of the algorithm is to consider the issue to address as a sparse regression problem, and to express the recorded multichannel signal as a linear combination of template shapes, by means of Overlapping Group LASSO (OGL). The optimization problem involved is solved in practice using the well-known ADMM algorithm, and can be accelerated using the method proposed in [39]. However, it is well known in the compressed sensing framework that a good sparse solution can only be obtained if there is a sufficiently good adequacy between the signal and the template shapes used in the dictionary (see e.g. [53] and references therein). Therefore, before applying the OGL, the recorded signal have to undergo a preliminary processing. Eventually, a final post-processing is then applied to determine the presence of pulses based on the OGL solution, $\hat{\mathbf{x}}_{GL}(r)$.

3.2. Intra-Cardiac ECG Preprocessing

The preprocessing suggested to make the data suited to a dictionary made of translated Gaussian shapes is the same processing used in the Pan-Tompkins algorithm for QRS detection in regular ECGs on each of the recorded channels separately [54]. More specifically, we apply on each channel separately a band-pass filter which keeps frequencies ranging from 5 Hz and up to 15 Hz. Then, we apply a numerical differentiation of the squared filtered channels, and eventually we apply a lowpass moving average filter. The main advantage of

such pre-processing step is that the resulting peaks are well suited to a dictionary of translated Gaussian shapes, thus increasing the suggested method's performance. The preprocessing's algorithm is summarized in Algorithm 1.

Algorithm 1 Pan-Tomkins based preprocessing

```

1: procedure PREPROCESSING( $\mathbf{y}$ )
2:   Set  $N_y$  = number of columns in  $\mathbf{y}$ 
3:   for  $n = 1 \dots N_y$  do
4:      $y_n = n$ -th column of  $\mathbf{y}$ 
5:      $a_n = BPF(y_n)$ 
6:      $b_n[k] = a_n[k] - a_n[k - 1], k \geq 2$ 
7:      $c_n = b_n^2$ 
8:      $\tilde{y}_n[k] = \frac{1}{50} \sum_{m=1}^{50} c_n[m]$ 
9:   end for
10:  return  $\tilde{\mathbf{y}}$ 
11: end procedure

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3.3. OGL Numerical Solution Using an Accelerated ADMM

Several approaches have been suggested for solving the OGL problem. For example, [55] suggests to duplicate columns of the dictionary appearing in more than one group, before optimizing the Group LASSO's cost function. However, this approach is not practical in the case of large dictionaries. In another contribution, [56] showed that the overlapping group case could be solved efficiently by approximating the proximal operator as a solution of an optimization sub-procedure that could be solved efficiently. However, this method still relies on a dictionary which needs to be fully stored in memory. In this paper, we extend the acceleration of the ADMM suggested in [38] for LASSO, which allows the processing of large signals to be approximated with large dictionaries, to the overlapping group LASSO. Though the regular group LASSO without overlapping can be solved efficiently using a variety of methods [38], the same

285 optimization problem with overlapping groups is much more challenging, and
 286 cannot be solved in a quick manner using projections. It was noted in [37]
 287 that the Alternated Direction Method of Multipliers (ADMM) could be used
 288 to solve the overlapping group LASSO. However, [37] used ADMM in a gen-
 289 eral consensus learning framework, and did not relate this particular case to
 290 LASSO. We thus rewrite below the ADMM algorithm for overlapping group
 291 LASSO. Additionally, we remark that this problem is homogeneous in the sense
 292 defined in [39]: for all $\alpha > 0$, if $\mathbf{x}_{GL}$ minimizes (2), then $\mathbf{x}_{GL}/\alpha$ minimizes
 293 $\frac{1}{2}\|\mathbf{y} - \alpha\mathbf{B}\mathbf{x}\|_2^2 + r\alpha \sum_{G \in \mathcal{G}} \|\mathbf{x}\|_{2,G}$. This last point is crucial for the acceleration
 294 proposed afterwards, since it implies that the matrices and vectors in (2) can
 295 be rescaled at will without a notable change in the solution. Consequently, the
 296 ADMM can be implemented efficiently, up to a rescaling of the variables.

297 Let us first rewrite (2) in a more suitable way for ADMM. We shall follow
 298 the formalism introduced in [37] for the consensus learning, and add equality
 299 constraints to enforce overlapping. Thus, (2) can be rewritten as

$$\begin{cases} \arg \min_{\mathbf{x}, \mathbf{z}_i, 1 \leq i \leq |\mathcal{G}|} \frac{1}{2}\|\mathbf{y} - \mathbf{B}\mathbf{x}\|_2^2 + r \sum_{i=1}^{|\mathcal{G}|} \|\mathbf{z}_i\|_2 \\ \text{such that } \mathbf{G}_i\mathbf{x} - \mathbf{z}_i = \mathbf{0}, \text{ for all } 1 \leq i \leq |\mathcal{G}| \end{cases}$$

300 where $\mathbf{G}_i$ is the matrix such that $\mathbf{G}_i\mathbf{x} = \mathbf{x}_{|G_i}$, where $\mathbf{x}_{|G_i}$ is the subvector
 301 derived from the coefficients whose indices are in the group G_i , and $|\mathcal{G}|$ denotes
 302 the overall number of groups. This is now in a form compatible with ADMM.
 303 We then define the augmented Lagrangian of the function to minimize as

$$\begin{aligned}
 L_\rho(\mathbf{x}, \mathbf{z}_1, \dots, \mathbf{z}_{|\mathcal{G}|}, \boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_{|\mathcal{G}|}) \\
 &= \frac{1}{2}\|\mathbf{y} - \mathbf{B}\mathbf{x}\|_2^2 + r \sum_{i=1}^{|\mathcal{G}|} \|\mathbf{z}_i\|_2 \\
 &+ \sum_{i=1}^{|\mathcal{G}|} \left(\boldsymbol{\lambda}_i^T (\mathbf{G}_i\mathbf{x} - \mathbf{z}_i) + \frac{\rho}{2} \|\mathbf{G}_i\mathbf{x} - \mathbf{z}_i\|_2^2 \right); \tag{3}
 \end{aligned}$$

Equation (3) can be rewritten in its scaled form, that is:

$$\begin{aligned}
& L_\rho(\mathbf{x}, \mathbf{z}_1, \dots, \mathbf{z}_{|\mathcal{G}|}, \boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_{|\mathcal{G}|}) \\
&= \frac{1}{2} \|\mathbf{y} - \mathbf{B}\mathbf{x}\|_2^2 + r \sum_{i=1}^{|\mathcal{G}|} \|\mathbf{z}_i\|_2 \\
&+ \sum_{i=1}^{|\mathcal{G}|} \left(\frac{\rho}{2} \left\| \mathbf{G}_i \mathbf{x} - \mathbf{z}_i + \frac{\boldsymbol{\lambda}_i}{\rho} \right\|_2^2 - \frac{\rho}{2} \left\| \frac{\boldsymbol{\lambda}_i}{\rho} \right\|_2^2 \right).
\end{aligned}$$

Now, defining $\mathbf{u}_i = \frac{\boldsymbol{\lambda}_i}{\rho}$, the steps of a scaled ADMM procedure can be written

as

$$\mathbf{x}^{(k+1)} = \arg \min_{\mathbf{x}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{B}\mathbf{x}\|_2^2 + \sum_{i=1}^{|\mathcal{G}|} \frac{\rho}{2} \left\| \mathbf{G}_i \mathbf{x} - \mathbf{z}_i^{(k)} + \mathbf{u}_i^{(k)} \right\|_2^2 \right\} \quad (4a)$$

$$\mathbf{z}_i^{(k+1)} = \arg \min_{\mathbf{z}_i} \left\{ r \|\mathbf{z}_i\|_2 + \frac{\rho}{2} \left\| \mathbf{G}_i \mathbf{x}^{(k+1)} - \mathbf{z}_i + \mathbf{u}_i^{(k)} \right\|_2^2 \right\}, \quad i = 1, \dots, |\mathcal{G}| \quad (4b)$$

$$\mathbf{u}_i^{(k+1)} = \mathbf{u}_i^{(k)} + \mathbf{G}_i \mathbf{x}^{(k+1)} - \mathbf{z}_i^{(k+1)}, \quad i = 1, \dots, |\mathcal{G}|. \quad (4c)$$

At this stage, it is important to notice that (4b) can be solved using a standard group soft thresholding, as shown later. Moreover, the $|\mathcal{G}|$ optimization problems in (4b), as well as the $|\mathcal{G}|$ updates in (4c), can be performed in parallel, since the variables $\mathbf{z}_i$ and $\mathbf{u}_i$ do not overlap. The main bottleneck of the method is in (4a). However, the function appearing in (4a) is differentiable with respect to $\mathbf{x}$. Computing the gradient with respect to $\mathbf{x}$ yields

$$\begin{aligned}
& \nabla_{\mathbf{x}} L_\rho(\mathbf{x}, \mathbf{z}_1, \dots, \mathbf{z}_{|\mathcal{G}|}, \boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_{|\mathcal{G}|}) \\
&= -\mathbf{B}^T(\mathbf{y} - \mathbf{B}\mathbf{x}) + \sum_{i=1}^{|\mathcal{G}|} \rho \mathbf{G}_i^T (\mathbf{G}_i \mathbf{x} - \mathbf{z}_i^{(k)} + \mathbf{u}_i^{(k)}).
\end{aligned} \quad (5)$$

Let us define $\mathbf{D} := \sum_{i=1}^{|\mathcal{G}|} \mathbf{G}_i^T \mathbf{G}_i = \mathbf{diag}(n_i)$, where n_i in the diagonal matrix of the right-hand side represents the overall number of occurrences of the i -th entry of $\mathbf{x}$ in the groups. Then, equating the gradient in (5) to zero gives

$$\mathbf{B}^T \mathbf{y} + \rho \sum_{i=1}^{|\mathcal{G}|} \mathbf{G}_i^T (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)}) = (\mathbf{B}^T \mathbf{B} + \rho \mathbf{D}) \mathbf{x}$$

Since all the entries of $\mathbf{x}$ appear at least once in the groups, the matrix $\mathbf{D}$ is invertible; consequently, the first update of the ADMM procedure is

$$\mathbf{x}^{(k+1)} = (\mathbf{D}^{-1}\mathbf{B}^T\mathbf{B} + \rho\mathbf{I})^{-1} \left(\mathbf{D}^{-1}\mathbf{B}^T\mathbf{y} + \rho \sum_{i=1}^{|\mathcal{G}|} \mathbf{D}^{-1}\mathbf{G}_i^T (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)}) \right). \quad (6)$$

The operation in (4b) is precisely the proximal operator, which, in the case of group LASSO, is the vector soft thresholding operation, as given in [37]:

$$\mathbf{z}_i^{(k+1)} = \max \left\{ 0, 1 - \frac{r}{\rho \|\mathbf{G}_i \mathbf{x}_i^{(k+1)} + \mathbf{u}_i^{(k)}\|_2} \right\} (\mathbf{G}_i \mathbf{x}_i^{(k+1)} + \mathbf{u}_i^{(k)}).$$

We summarize the standard ADMM applied to solve the OGL problem in algorithm 2. It should be noted that the objective described in the OGL optimization is a sum of convex terms. Consequently, the existence of a solution is guaranteed. Moreover, the convergence of ADMM does not depend on the equality constraints, but on mild conditions on the function to minimize. More specifically, if both the error and penalty terms are closed, proper convex functions, and if the unaugmented Lagrangian of the function to minimize has a saddle point (which is the case in practice for OGL and, more generally, for LASSO and its variants), then the iterates $\mathbf{x}^{(k)}, \mathbf{z}_1^{(k)}, \dots, \mathbf{z}_{|\mathcal{G}|}^{(k)}$ converge to the feasible set, and the objective function tends to its minimal value (see Appendix A in [37]). Moreover, under additional conditions, it can be shown that the iterates converge to optimal values, as seen in [57] and references therein. The rate of convergence for group LASSO (*a fortiori* for OGL) is still a topic under investigation [58, 59], and is not in the scope of this contribution.

In the standard ADMM iteration, it is clear that the main bottleneck is the computation of $\mathbf{x}^{(k+1)}$. However, equation (4a) shares similarities with the regular ADMM update. It is therefore possible to use the acceleration scheme presented in [39] for the fast computation of (6). Assuming that $\text{Sp}(\mathbf{B}^T\mathbf{B}) < 1$, where $\text{Sp}(\mathbf{A})$ denotes the spectral radius of a square matrix $\mathbf{A}$, (6) can be

Algorithm 2 Standard ADMM for OGL

```

1: procedure ADMM( $\tilde{\mathbf{y}}, \mathbf{B}, r, \rho, tol_x, \mathbf{G}_1, \dots, \mathbf{G}_{|\mathcal{G}|}$ )
2:    $k = 1, \mathbf{x}^0 = \mathbf{0}$ 
3:   Compute  $\mathbf{D} := \sum_{i=1}^{|\mathcal{G}|} \mathbf{G}_i^T \mathbf{G}_i$ 
4:   while  $\|\mathbf{x}^k - \mathbf{x}^{k-1}\|_2 < tol_x$  do
5:      $k \leftarrow k + 1$ 
6:      $\mathbf{x}^{(k+1)} = (\mathbf{D}^{-1} \mathbf{B}^T \mathbf{B} + \rho \mathbf{I})^{-1} (\mathbf{D}^{-1} \mathbf{B}^T \mathbf{y} + \rho(\bar{\mathbf{z}}^{(k)} - \bar{\mathbf{u}}^{(k)}))$   $\triangleright$  This step
       can be accelerated
7:      $\mathbf{z}_i^{(k+1)} = \max \left\{ 0, 1 - \frac{r}{\rho \|\mathbf{G}_i \mathbf{x}_i^{(k+1)} + \mathbf{u}_i^{(k)}\|_2} \right\} (\mathbf{G}_i \mathbf{x}_i^{(k+1)} + \mathbf{u}_i^{(k)}), i =$ 
        $1 \dots |\mathcal{G}|.$ 
8:      $\mathbf{u}_i^{(k+1)} = \mathbf{u}_i^{(k)} + \mathbf{G}_i \mathbf{x}^{(k+1)} - \mathbf{z}_i^{(k+1)}, i = 1, \dots, |\mathcal{G}|$ 
9:   end while
10:  return  $\hat{\mathbf{x}}_{GL}(r) = \mathbf{x}^k$ 
11: end procedure

```

336 rewritten as

$$\mathbf{x}^{(k+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\rho^n} (\mathbf{D}^{-1} \mathbf{B}^T \mathbf{B})^n \left(\frac{1}{\rho} \mathbf{D}^{-1} \mathbf{B}^T \mathbf{y} + \sum_{i=1}^{|\mathcal{G}|} \mathbf{D}^{-1} \mathbf{G}_i^T (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)}) \right). \quad (7)$$

In (7), all the multiplications by $\mathbf{B}$ and $\mathbf{B}^T$ can be computed efficiently, using the acceleration scheme presented in [39]. Moreover, we can observe that $\mathbf{G}_i \mathbf{z}_i^{(k)}$ is a vector of the same dimensions as $\mathbf{x}$, whose m -th entry equals 0 if $m \notin G_i$, and equals $\mathbf{z}_{i,m}^{(k)}$ otherwise. Consequently, $\sum_{i=1}^{|\mathcal{G}|} \mathbf{G}_i^T \mathbf{z}_i^{(k)}$ is a vector whose m -th entry is equal to the sum of all the entries of the vectors $\mathbf{z}_i^{(k)}$ mapped to the m -th position by any of the $\mathbf{G}_i$. Therefore, we can write

$$\sum_{i=1}^{|\mathcal{G}|} \mathbf{D}^{-1} \mathbf{G}_i^T (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)}) = \bar{\mathbf{z}}^{(k)} - \bar{\mathbf{u}}^{(k)},$$

337 where $\bar{\mathbf{u}}^{(k)}$ and $\bar{\mathbf{z}}^{(k)}$ are vectors whose m -th entry is the average of the represen-
338 tative variables of all the entries of the vectors $\mathbf{u}_i^{(k)}$ and $\mathbf{z}_i^{(k)}$ mapped to the m -th

position by any of the $\mathbf{G}_i$. These steps can be computed quickly, since the matrix $\mathbf{D}$ is diagonal and is known from the beginning, and the calculation of $\bar{\mathbf{u}}^{(k)}$ and $\bar{\mathbf{z}}^{(k)}$ can be computed iteratively. It is therefore possible to summarize the fast update of $\mathbf{x}^{(k+1)}$ in Algorithm 3. It can be noted that the matrices $\mathbf{G}_i$ are projection matrices, thus multiplications by $\mathbf{G}_i^T$ consist in fact in selecting the relevant coefficients from the vector used in the multiplication. Consequently, the computational burden of these operations is negligible, compared to the computation of $\mathbf{c}_n$ and $\mathbf{d}_n$ (as defined in Algorithm 3). Similarly, the multiplication by $\mathbf{D}^{-1}$ is not computationally cumbersome, since $\mathbf{D}^{-1}$ is diagonal. Thus the complexity of one ADMM iteration is determined by the complexity of the computation of $\mathbf{c}_n$ and $\mathbf{d}_n$. More specifically, using the notations detailed in Algorithm 3, we get that the algorithmic complexity related to one update of $\mathbf{x}^{(k+1)}$ is $O(N_{max}p(Np)\log_2(Np))$. Since the updates of $\mathbf{z}_i^{(k+1)}$ and $\mathbf{u}_i^{(k+1)}$ have both a linear complexity in N , the complexity of one ADMM update is equal to the complexity of one update of $\mathbf{x}^{(k+1)}$.

3.4. Post-processing for Pulse Arrival Estimation

Assuming that we have obtained a candidate regressor $\hat{\mathbf{x}}_{GL}(r)$, we now need to extract from $\hat{\mathbf{x}}_{GL}(r)$ estimators of the arrival times associated with each electrical pulse. However, the observed array of signals is not created using the chosen dictionary $\mathbf{B}$. Indeed, the shape of the observed pulses can change substantially from one channel to another and may also be slowly time-varying within each channel. Since $\mathbf{B}$ only approximates the recorded ECG, it is likely that, when a pulse occurs, the OGL will activate several consecutive groups in order to provide a good regressor, and that a few coefficients inside the groups will represent the main waveform, whereas the remaining coefficients will be much smaller and just serve to refine the regression. Thus, an individual pulse on one channel will not be estimated with a single given shape, but rather by a combination of shapes found in consecutive groups. Furthermore, since $\hat{\mathbf{x}}_{GL}(r)$ is the solution of the OGL, individual shapes inside a given group are either all active or inactive. Both facts can become an issue if our aim is estimating

Algorithm 3 Fast computation of $\mathbf{x}^{(k+1)}$ using Fast Fourier Transforms (FFTs)
 (see [39] for details)

```

1: procedure FAST COMPUTATION( $\mathbf{B}, \mathbf{y}, N_{max}, \mathbf{D}, \mathbf{G}_i, \mathbf{z}_i^{(k)}, \mathbf{u}_i^{(k)}$ )
2:   Set  $\mathbf{B} = [\mathbf{B}_1 \ \mathbf{B}_2 \ \dots \ \mathbf{B}_p]$ , where  $\mathbf{B}_s$  is the submatrix generated by the
      $k$ -th translated template shape  $\mathbf{b}_s$ .
3:   Compute  $\mathbf{a}^{(0)} = \frac{1}{\rho} \mathbf{D}^{-1} \mathbf{B}^T \mathbf{y} + \sum_{i=1}^{|\mathcal{G}|} \mathbf{D}^{-1} \mathbf{G}_i^T (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)})$ 
4:   Set  $Np$  = number of entries of  $\mathbf{y}$ .
5:   for  $n = 1 \dots N_{max}$  do
6:     Set  $\mathbf{a}^{(n-1)} = \begin{bmatrix} \mathbf{a}_1^{(n-1)} \\ \mathbf{a}_2^{(n-1)} \\ \vdots \\ \mathbf{a}_p^{(n-1)} \end{bmatrix}$ , such that  $\mathbf{B} \mathbf{a}^{(n-1)} = \sum_{s=1}^p \mathbf{B}_s \mathbf{a}_s^{(n-1)}$ 
7:     Compute  $\mathbf{c}_n = \sum_{s=1}^p \text{IFFT}_{Np}(\text{FFT}_{Np}(\mathbf{b}_s) \times \text{FFT}_{Np}(\mathbf{a}_s^{(n-1)}))$ 
8:     Compute  $\mathbf{d}_n = \mathbf{B}^T \mathbf{c}_n = \begin{bmatrix} \mathbf{B}_1^T \mathbf{c}_n \\ \mathbf{B}_2^T \mathbf{c}_n \\ \vdots \\ \mathbf{B}_p^T \mathbf{c}_n \end{bmatrix}$ , where  $\mathbf{B}_s^T \mathbf{c}_n =$  The cross-correlation  
is also computed ef-  
ficiently using Fast  
Fourier Transforms
9:     Compute  $\mathbf{a}_n = -\frac{1}{\rho} \mathbf{D}^{-1} \mathbf{d}_n$ 
10:   end for
11:   return  $\mathbf{x}^{(k+1)} = \sum_{n=0}^{N_{max}} \mathbf{a}^{(n)}$ 
12: end procedure

```

369 the pulse arrival times. In order to solve this problem, we introduce a post-
 370 processing step adapted to the OGL particularities.

Given that group G associated to k consecutive sampling times $\{n, n + 1, \dots, n + k - 1\}$, we first sum for each channel the coefficients of $\hat{\mathbf{x}}_{GL}(r)$ in G related to the same sampling time. This operation provides M column vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M$ of size k , with each one relating to a single channel. Due to the theoretical properties of the Grouped LASSO, the $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M$ are either equal to 0 altogether, or fully non-zero altogether. A threshold T is then applied to each $\mathbf{v}_j$ in order to cancel its smallest coefficients, which are related to the refinement of the signal estimation and not to the localization of the pulses themselves. The output of this thresholding operation for the i -th vector is the vector $\tilde{\mathbf{v}}_i$, whose k -th component is

$$\tilde{v}_{i,j} = \begin{cases} v_{i,j}, & \text{if } v_{i,j} \geq T; \\ 0, & \text{if } v_{i,j} < T. \end{cases}$$

371 If the vectors $\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_M$ are different from 0 after thresholding, then we
 372 can relate them to arrival times of electrical pulses.

373 After this post-processing, two estimation strategies can be envisaged for
 374 the arrival times: 1) a time estimation per individual channel, or 2) a global,
 375 consensus-based, time estimation. The time estimation for each channel indi-
 376 vidually is fairly straightforward: given the $\tilde{\mathbf{v}}_k$'s, the arrival times are estimated
 377 as the inner product

$$\hat{m}_{n,i} = \mathbf{t}_n \cdot \frac{\tilde{\mathbf{v}}_i}{\|\tilde{\mathbf{v}}_i\|_1}, \quad i = 1 \dots M,$$

378 where the vector $\mathbf{t}_n$ is defined as $\mathbf{t}_n = [n, n + 1, \dots, n + k - 1]$ and $\|\cdot\|_1$
 379 denotes the ℓ_1 norm. This approach (denoted by *individual estimation* later on)
 380 provides a precise estimation of the pulses' arrival times, though in this process
 381 we lose the dependency between the channels.

382 Another, more interesting approach, is to rely on the δ -synchronicity as-
 383 sumed for the multichannel signal at hand. Instead of estimating the arrival
 384 times individually, we choose δ -neighborhoods $V_k(\delta)$ by consensus selection.

385 More precisely, a neighborhood $V_k(\delta)$ is selected as an area where an electri-
 386 cal pulse has occurred if a user-defined percentage of the thresholded vectors
 387 $(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_M)$ are different from zero. For instance, the most straightfor-
 388 ward approach is to follow a majority voting scheme: given M channels, if at
 389 least $M/2$ vectors $\tilde{\mathbf{v}}_i$ are not null then an activation is detected; otherwise, we
 390 consider that no activation has occurred. However, as discussed in Section 4,
 391 requiring a smaller number of simultaneously non-null vectors $\tilde{\mathbf{v}}_i$ may be desir-
 392 able. In any case, if we decide that an electrical pulse has occurred, then we
 393 simply estimate the arrival time as the midpoint of the selected group. Such
 394 an approach (denoted by *consensus selection* later on) is well suited with OGL
 395 regression, since the active groups can be tailored to represent both the window
 396 size δ and the correlation between channels. We may lose in precision regarding
 397 the individual time estimation, since we replace the problem of point estima-
 398 tion by a selection of neighborhoods including the actual times; but consensus
 399 selection can help to discard regions where false activations occur, as well as to
 400 select regions of noisy channels where pulse reconstruction is made difficult. We
 401 [summarize the consensus selection approach in Algorithm 4.](#)

402 4. Results and Discussion

403 We present the results obtained on synthetic and real data. Synthetic data
 404 were used to investigate the dependency relationship between all the paramet-
 405 ers appearing in the proposed approach, while for results on real data these
 406 parameters were set up manually. As discussed before, in this paper the groups
 407 are always composed of k consecutive sampling points with a certain overlap
 408 with respect to the previous group. In this section, the overlap between the
 409 groups is given as a percentage, since all the groups are of the same size. More
 410 specifically, if $|G| = k$ denotes the number of elements in each group and q de-
 411 notes the user-defined overlap percentage, the overlapping number of elements
 412 is given by $[qk/100]$, where $[\cdot]$ denotes the rounding to the nearest integer.

Algorithm 4 Post-processing for joint peak localization by consensus

1: **procedure** POST-PROCESSING($\hat{\mathbf{x}}_{GL}, T, \text{voting threshold}$)

2: Define for all $n = 0 \dots N - 1$:

$$\mathcal{A}_n = \{0 \leq m \leq Np ; \exists G \in \mathcal{G} \text{ s.t. } \{m, n\} \in G\}$$

3: Define for all $k = 1 \dots M$:

$$\mathbf{v}_k[n] = \sum_{m \in \mathcal{A}_n} \mathbf{x}_{GL}[m], \quad n = 0 \dots N - 1.$$

4: **for** $k = 1 \dots M$ **do** ▷ Thresholding

5: Update $\tilde{\mathbf{v}}_k[n] = \begin{cases} \mathbf{v}_k[n] & \text{if } \mathbf{v}_k[n] > T \\ 0 & \text{otherwise} \end{cases}, \quad n = 0 \dots N - 1$

6: **end for**

7: Define $\mathbf{s} = [\]$, $\mathbf{e} = [\]$, $p[0] = 0$

8: **for** $n = 1 \dots N - 1$ **do** ▷ Consensus selection

9: Compute

$$p[n] = \begin{cases} 1 & \text{if } \sum_{k=1}^M 1(\tilde{\mathbf{v}}_k[n] > 0) > \text{voting threshold} \\ 0 & \text{otherwise} \end{cases}$$

10: **if** $p[n] = 1$ and $p[n - 1] = 0$ **then**

11: $\mathbf{s} = [\mathbf{s} \ n]$

12: **else if** $p[n] = 0$ and $p[n - 1] = 1$ **then**

13: $\mathbf{e} = [\mathbf{e} \ n]$

14: **end if**

15: **end for**

16: **return** $\frac{\mathbf{e} + \mathbf{s}}{2}$

17: **end procedure**

413 Note that, unlike external ECGs, where there are many public annotated
 414 databases, having access to annotated intracardiac ECGs (a.k.a. EGMs) is
 415 rather uncommon. Indeed, freely available databases of annotated EGMs do
 416 not exist as far as we know,² and currently we do not have access to annotated
 417 EGMs either. For this reason, researchers typically validate their methods first
 418 using synthetic data (i.e., simulations) in order to obtain numerical results and
 419 then apply them to real EGMs, where only qualitative results can be provided.
 420 For example, this approach is followed in [31, 32], where they use first a nu-
 421 merical model to simulate a propagation pattern and test their method, and
 422 then they provide a qualitative analysis of their results on real data. This is the
 423 approach that we have followed here: numerically validating the performance
 424 of our approach on a synthetic example and then testing it on real data, where
 425 we provide qualitative results that confirm its good performance.

426 4.1. Synthetic Data and Performance Evaluation

427 In our simulation, we presented the optimal case, when synthetic signals
 428 are generated from shapes included in the dictionary $\mathbf{B}$, based on the model
 429 described in (1). More specifically, in (1), the number of channels is taken
 430 equal to $M = 10$, while the number of sampling points is set to $N = 100$. For
 431 all the channels, we simulate the arrival times of each electrical pulse $m_{k,i}$ as a
 432 sample path of a renewal process, such that the inter-arrival times $m_{k+1,i} - m_{k,i}$
 433 are independent and identically distributed with a common normal distribution
 434 $N(30, 0.25)$. These selected values for the expectation and standard deviation
 435 allow for slight random shifts between pulses in each channel, while keeping the
 436 probability of large discrepancies between channels extremely low. The shapes

²The only public EGM database that we are aware of is the Intracardiac Atrial Fibrillation Database, which is available through Physionet [60] (<https://physionet.org/content/iafdb/1.0.0/>). This database consists of endocardial recordings from the right atria of 8 patients in atrial fibrillation or flutter which are not annotated by cardiologists. The creators of the database have annotated the data using the simple SQRS algorithm [61], but this cannot be considered as the ground truth for performance evaluation.

$\Phi_{k,i}$ for each channel are chosen randomly, based on a uniform draw, between
 ten normalized Gaussian curves, whose translations define the dictionary $\mathbf{B}$.
 The amplitudes $a_{k,i}$ are drawn based on a uniform distribution on $[30, 40]$. A
 Gaussian noise with standard deviation 0.1 is then added to all the channels.
 Figure 3 illustrates the behavior of such multichannel simulated signals. For
 illustrative purposes, we present in Figure 4 the comparison between the actual
 times and the estimated times for the signals generated in Figure 3 using a
 sparsity parameter $r = 0.01$, and groups of 5 consecutive sampling points with
 a 25% overlap (i.e., one sample overlap between consecutive groups). We observe
 that the proposed approach provides a quite satisfactory estimation of the arrival
 times in this setting.

In the following section we investigate the performance of the proposed OGL
 method with respect to its key parameters. In order to numerically evaluate the
 efficiency of our approach, we use the following criterion. Given any set of
 integers A from $\{0, 1, \dots, N-1\}$, we define the augmented sets $A^{(k)}$ as follows

$$A^{(k)} = \bigcup_{i \in A} \{\max(0, i-k), \dots, \min(i+k, N-1)\}$$

Thus, two sets of integers A and B can be compared by means of the following
 criterion:

$$D(A, B) = \frac{1}{N} \sum_{k=0}^{N-1} d(A^{(k)}, B^{(k)}), \quad (8)$$

where $d(A^{(k)}, B^{(k)}) = 1 - \frac{|A^{(k)} \cap B^{(k)}|}{|A^{(k)} \cup B^{(k)}|}$ is the Jaccard's distance between the
 sets $A^{(k)}$ and $B^{(k)}$, $|A|$ denotes the number of integers in A and we set $A^{(0)} =$
 $A, B^{(0)} = B$. Note that $D(A, B) = 0$ if and only if $A = B$. Furthermore, (8)
 reflects inner variability between time occurrences: if A and B are different but
 share a similar pattern (e.g. the integers of B are close to the ones in A), the
 obtained distance $D(A, B)$ will be small. With these considerations in mind, the
 efficiency of the proposed approach is evaluated by computing $\sum_{k=1}^M D(A^k, B^k)$,
 where A^k and B^k denote the sets of true and estimated arrival times for the
 k -th channel, respectively.

463 In order to show that the proposed distance in (8) is able to discriminate
 464 good and bad QRS detectors, we perform a toy experiment. On the one hand,
 465 we generate a synthetic list of activation times,

$$A_I = \{1, P + 1, 2P + 1, \dots, RP + 1\},$$

466 where $R = \lfloor \frac{N}{P} \rfloor$ is the number of true activations in this example. On the other
 467 hand, we generate an initial noisy list of activation times by introducing a lag
 468 L ($0 \leq L \leq P - 1$), so that

$$A_N = \{1, P + L + 1, 2P + L + 1, \dots, RP + L + 1\}.$$

469 Note that the sample associated to the r -th ideal activation ($1 \leq r \leq R$)
 470 in this case is $A_I(r) = rP + 1$, whereas for the r -th noisy activation it is
 471 $A_N(r) = rP + L + 1$, but the same number of activations are present in both
 472 sequences. Finally, we incorporate the possibility of having non-detected acti-
 473 vations, i.e., $A_N(r) = rP + L + 1$ is removed from A_N with probability P_{ND}
 474 ($0 \leq P_{ND} \leq 1$) independently for each value of r . Therefore, the expected
 475 number of activations in A_N is $(1 - P_{ND})R$.

476 For this example, we set $P = 200$, $K = \frac{P}{2} = 100$, $N = 10^4$ (i.e., $R = \lfloor \frac{N}{P} \rfloor =$
 477 50), $0 \leq L \leq \frac{P}{2} = 100$, and $0 \leq P_{ND} \leq 1$. Figures 5a and 5b show the evolution
 478 of $d(A_I^{(k)}, A_N^{(k)})$ as a function of k for different values of the lag (setting $P_{ND} = 0$)
 479 and the non-detection probability (setting $L = 1$) respectively. In both cases,
 480 a substantial difference can be observed among the difference curves, with a
 481 much slower decay of $d(A_I^{(k)}, A_N^{(k)})$ as the lag and the non-detection probability
 482 are increased, thus leading to larger values of $D(A_I, A_N)$. Indeed, note that
 483 $D(A_I, A_N)$ essentially corresponds to the area under the curve $d(A_I^{(k)}, A_N^{(k)})$,
 484 which is a well-known performance measure used to compare several detectors
 485 in statistical signal processing and machine learning. A slower decay means a
 486 larger area under the curve, and thus a larger distance between the list of ideal
 487 and noisy activations (i.e., a worse performance). This behaviour is confirmed
 488 by Figures 6a and 6b, which display $D(A_I, A_N)$ as a function of L and P_{ND}
 489 respectively. On the one hand, from Figure 6a it can be seen that $D(A_I, A_N)$

490 has an increase of the type $1 - \exp(-\gamma x)$ with respect to L for a fixed value of
 491 P_{ND} (with a higher speed the smaller the value of P_{ND}). On the other hand,
 492 from Figure 6b it can be seen that $D(A_I, A_N)$ increases linearly with respect
 493 to P_{ND} for a fixed value of L (with a higher slope the smaller the value of L).
 494 In both cases, $D(A_I, A_N)$ is a monotonically increasing function, implying that
 495 higher values of both L and P_{ND} always lead to larger values of $D(A_I, A_N)$ as
 496 expected.

497 4.2. Results on Synthetic Data

498 The choice of the threshold parameter used is important to refine the so-
 499 lution obtained by means of OGL. On one hand, setting too large a threshold
 500 will artificially set some coefficients of the regressor to 0, yielding undetected
 501 pulses. On the other hand, a small threshold can lead to false activations and
 502 therefore to false pulse detection. With these considerations in mind, it is clear
 503 that a good threshold parameter is signal dependent, and is strongly related
 504 to the noise variance. We empirically investigated the influence of the thresh-
 505 old parameter on the distance D described above between the actual simulated
 506 times and the estimated arrival times. We performed 100 simulations with a
 507 sparsity parameter $r = 0.01$, a group size of 25 samples, an overlap between
 508 groups of 25%, and a noise variance of 0.1. Figure 7 illustrates the influence of
 509 the threshold when using the individual estimation strategy, and Figure 8 dis-
 510 plays the achieved results with the consensus selection strategy. In both cases,
 511 the obtained curves share strong similarities, illustrating the intuitive idea pre-
 512 sented above on the relationship between the threshold and the error. Indeed,
 513 both show that good results are attained when selecting a threshold around the
 514 true noise variance (0.1). However, note that the performance of the consensus
 515 strategy is much more robust with respect to the threshold choice, remaining
 516 stable for a larger range of threshold values. Consequently, in the rest of the
 517 paper we set the threshold equal to the noise variance. The automatic choice of
 518 the threshold and the sparsity parameter for peak localization is out of the scope
 519 of the present paper. However, note that the shortcomings of cross-validation

for the latter task, and a possible way to select it automatically in a related framework, have been illustrated in [62].

In a second experiment, we estimate the arrival times for each multichannel signal by means of Group LASSO, investigating the influence on the performance for three different parameters: the sparsity parameter r , the percentage of overlapping between groups, and the parameter δ used to define δ -synchronicity. For a given choice of these three values, we repeat the experiment 100 times for different generated signals. We thus compare the estimated arrival times to their actual counterparts. The summary of the quantitative performance of the proposed approach on the synthetic data for these three parameters is presented in Table 2. The obtained results emphasize two important points. The first point to consider is the choice of the sparsity parameter. Indeed, a too small value of r provides a less sparse solution for OGL, yielding less precision in the estimation of the pulses' arrival times. Furthermore, it shows that adding some overlap between the groups improves the precision of the reconstruction, though too much overlap does not improve the performance of the algorithm anymore. The results obtained on these two simulations validate the principles of the proposed approach.

4.3. Results on Real Data

In this section, we show two applications of the proposed method on real ECG data provided by Dr. Angel Arenal's team from Madrid's *Hospital Universitario Gregorio Marañón* (Spain). The signals correspond to patients with sustained atrial fibrillation (AF) and were obtained during heart surgery. They can be divided into two groups: artificially induced sinus rhythm and AF signals. All of the recordings were performed using a lasso catheter directly in contact with the heart muscle, which provides $M = 10$ bipolar intracardiac ECGs (i.e., channels) with a sampling frequency of $f_s = 1/T_s = 977$ Hz. As mentioned above, the data are not annotated, so we rely on qualitative results (histograms of interarrival times for the LASSO, GL and OGL approaches) to show that the performance of the OGL is better than the performance of previous approaches

	$r = 0.0001$			$r = 0.001$			$r = 0.01$		
	25%	50%	75%	25%	50%	75%	25%	50%	75%
$\delta = 10$									
mean	0.0252	0.0220	0.0222	0.0237	0.0248	0.0207	0.0237	0.0218	0.0225
std	0.0144	0.0139	0.0136	0.0162	0.0152	0.0135	0.0133	0.0130	0.0138
$\delta = 25$									
mean	0.0221	0.0218	0.0250	0.0214	0.0221	0.0218	0.0230	0.0219	0.0219
std	0.0129	0.0139	0.0147	0.0125	0.0138	0.0143	0.0150	0.0145	0.0150
$\delta = 50$									
mean	0.0268	0.0238	0.0217	0.0228	0.0250	0.0251	0.0230	0.0208	0.0200
std	0.0141	0.0141	0.0140	0.0128	0.0150	0.0148	0.0147	0.0132	0.0120
$\delta = 75$									
mean	0.0245	0.0238	0.0220	0.0208	0.0211	0.0205	0.0213	0.0244	0.0241
std	0.0128	0.0130	0.0125	0.0156	0.0138	0.0130	0.0130	0.0139	0.0140

Table 2: Average and standard deviation values of the criterion D for different values of r , δ and the groups' overlap.

550 based on the LASSO and GL.

551 We used, for the presented results, a dictionary made of 10 translated Gaus-
552 sian shapes, with standard deviations ranging from 1 to 4. For the OGL settings,
553 we choose each group's size equal to 100, which corresponds approximately to
554 100 ms, that is about half the physiological refractory period of the heart mus-
555 cle. The overlap between the created groups was set to 50 %, and the sparsity
556 parameter was chosen equal to $r = 0.001$. Ten iterations of the proposed algo-
557 rithm were applied. For the consensus-based selection approach, the number of
558 minimal active channels for activation was set equal to 5 for the sinus rhythm,
559 and to 3 for AF data. Lowering the number of active channels for AF charac-
560 terizes the lack of synchronicity between channels occurring in atrial fibrillation
561 episodes. Note that this requires knowing in advance whether an AF episode is
562 present or not. In practice, this is often the case in clinical settings. Otherwise,
563 a simple frequency approach can be used to discriminate between sinus rhythm
564 and AF and set the most appropriate parameters [63, 64].

565 We first present the results obtained for sinus rhythm. Point estimates of
566 arrival times are displayed for the first 6000 sampling times in Figures 9 and 10
567 for consensus selection and individual estimation, respectively. For the sake of
568 visibility, we chose to display only the first two channels of the intracardiac
569 recording. However, the full graph with all the channels is displayed in the
570 supplementary material. Regions where electrical pulses likely occur (based on
571 the groups size chosen) are presented for the first 6000 points of all the channels
572 in Figure 11 for consensus selection, and in Figure 12 for individual estimation.
573 Since overlapping is allowed for OGL, these regions are displayed in light red,
574 whereas the darker the shades of red are, the more the selected regions overlap.
575 For the chosen setting, no overlap appears in Figure 11. The good performance
576 of the OGL is confirmed by the histograms shown in Figure 13. Note how the
577 OGL has a clear peak around an interarrival time ≈ 1.3 s, which corresponds to
578 ≈ 46 bpm (a reasonable value for a sedated patient in the operating room), with
579 just a few spurious values above and below. On the other hand, the interarrival
580 times recovered by LASSO also show a large peak around an interarrival time

581 ≈ 1.3 s, but a second large peak (almost as large as the previous one) appears
 582 at very low interarrival times (0–0.2 s). This indicates that LASSO is detecting
 583 many spurious consecutive peaks as valid heartbeats; in other words, it intro-
 584 duces many false alarms. Finally, GL attains the worst performance, with many
 585 spurious activations that lead to a single peak in the very low interarrival time
 586 range (0–0.2 s). This is due to the fact that GL “forces” activation whenever a
 587 single channel is active, thus detecting many spurious peaks.

588 Figures 14, 15, 16, and 17 illustrate the results for an atrial fibrillation
 589 episode. Similarly, we decided to present the results obtained in the full signal
 590 and all the channels in the supplementary material. Once more, the better
 591 performance of the OGL is confirmed by the histograms shown in Figure 18. As
 592 before, OGL provides the best results (with a single peak around interarrival
 593 times ≈ 0.15 s, i.e., ≈ 400 bpm), followed by LASSO (which has an additional
 594 large peak in the range 0–0.1 s) and GL (which has a single peak at interarrival
 595 times in the range 0–0.1 s). Overall, this suggests that the proposed OGL
 596 approach is doing a good job in detecting the true ECG pulses and introducing
 597 very few spurious detections.

598 4.4. Discussion

599 One important aspect of any compressive sensing based approach is the
 600 choice of the dictionary. Indeed, sparse reconstruction and consistency in se-
 601 lection are attainable if the dictionary’s shapes are well suited to the signal at
 602 hand and if conditions on the shapes’ correlations (e.g., as described in [65]) are
 603 met. In the case of intracardiac ECGs, since the data at hand is not annotated,
 604 the choice of Gaussian shapes is motivated by the output of the Pan-Tompkins
 605 filtering (see Figure 19), which provides shapes with one single-mode around
 606 ECG pulses. This choice, though suboptimal, provides very satisfactory results
 607 in practice. We also note that other shapes (such as those appearing in the sup-
 608plementary material) are also well suited for the pulses’ localization. However,
 609 the automatic derivation of the dictionary’s shapes, either by using dedicated
 610 algorithms [66] or by learning them from annotated templates as in [51], could

611 help to further circumvent the preprocessing step, and should be investigated
612 in future contributions.

613 We note, from an overall view on the results, that the proposed algorithm
614 for OGL regression succeeds in selecting parts of the signal which most likely in-
615 clude electrical pulses. By comparing Figure 11 with Figure 12, it can be easily
616 deduced that a consensus selection approach for intracardiac ECG pulse local-
617 ization outperforms an individual channel based approach, and this in several
618 aspects. Firstly, we can observe that the consensus approach relies on the posi-
619 tive vote of a user-defined percentage of all the channels regarding the selection
620 of active groups. Therefore, pulse localization by consensus is more robust to all
621 noise artifacts resembling electrical pulses, unrelated to the electrical activity
622 of the heart’s muscle but rather related to unwanted electrodes’ movements.
623 As seen in Figure 9, these artifacts are not selected, since they only occur on
624 few electrodes at a time, whereas an approach based on individual channels
625 (see Figure 10) tends to mistakenly consider these artifacts as true pulses. Fur-
626 thermore, the consensus approach allows selection of areas where the low SNR
627 cripples the pulse localization for some of the channels, as seen in Figures 9
628 and 11. On the other hand, processing channels with low SNRs individually, as
629 seen in Figure 10, fails to effectively address the pulses’ localization problem.

630 Another distinguishable point is that consensus based selection is more suited
631 to physiological constraints. Indeed, taking the existing correlation between the
632 channels into account, both in the OGL step and in the selection, provides
633 a localization which is physiologically sound. On the contrary, as seen from
634 Figure 12, by processing each channel independently we may select overlapping
635 or close regions, in opposition to the refractory period characteristic of ECG
636 signals.

637 From the global results appearing in the supplementary material for sinus
638 data, we can see that the consensus approach fails to select areas where most of
639 the channels have a low SNR. Three possible solutions to address this shortcom-
640 ing are, from the most time-consuming to the fastest: to increase the number of
641 iterations of the OGL algorithm, to lower the threshold used to discard the small

642 coefficients or to lower the number of positive votes required for selection. The
 643 last two approaches are similar to the strategy followed by the Pan-Tompkins
 644 algorithm, where the used thresholds are adaptively lowered when no pulses are
 645 detected in large parts of ECG signals. Indeed, lowering the number of positive
 646 votes from 5 to 3 allows to locate all the pulses, but introduces false detections
 647 in other parts of the signals. Thus, it is recommended to separately process
 648 regions where pulses should be detected with a lower number of positive votes,
 649 based on the standard physiological constraints.

650 Atrial fibrillation signals are much more complex to analyze, both due to a
 651 lower SNR and to less synchronicity between channels. Nevertheless, we can see
 652 from Figures 14 and 16 that OGL with consensus selection performs relatively
 653 well in pulse localization, and performs much better than individual channel
 654 processing as seen from the comparison with Figure 17.

655 The proposed approach is as quick as expected (one minute of 10 channels
 656 was processed in 30 minutes on an entry-budget computer with an i3 processor
 657 and 4 GB of RAM), though further investigation should be done to decrease
 658 the processing time and make it more suited to real-time or quasi-real-time ap-
 659 plications. However, following [37], we note that the steps of OGL can be easily
 660 parallelized. Such an improvement can easily reduce the computation times,
 661 but is out of the scope of the present paper. It is clear that OGL has a more
 662 complex formulation than regular LASSO (tested by the authors in [26]) or reg-
 663 ular group LASSO without overlaps (similarly tested in [30]), and therefore the
 664 three approaches are worthy of comparison. When applying ADMM with the
 665 same settings for LASSO, Group LASSO and OGL, we found that the additional
 666 computational cost required by OGL is rather marginal, although it is clear that
 667 LASSO and Group LASSO without overlap can be made quicker using proximal
 668 gradient. Regarding their performances, we present in Figures 20 and 21 results
 669 obtained with similar settings for the regular LASSO and group LASSO without
 670 overlaps on the sinus rhythm data. The details of both results are interesting.
 671 On the one hand, LASSO performs well in pulse localization, since the selected
 672 areas are close to the ones obtained by OGL. However, several areas are selected

673 more than once. This fact illustrates the well known difficulties of using LASSO
 674 for pulse localization when the dictionary is composed of translated waveforms.
 675 Indeed, in that case the required theoretical conditions for sparse reconstruction
 676 (e.g., see [65]) are not met. Though further post-processing (as done in [26])
 677 might help to reduce the number of selected areas, the removal of multiply se-
 678 lected areas will remain arbitrary, and we see that under the same conditions the
 679 proposed method performs better than ordinary LASSO. On the other hand, as
 680 previously observed in [30], ordinary group LASSO without overlaps performs
 681 rather poorly for pulse localization. This can be interpreted as follows: one im-
 682 portant theoretical property of group LASSO is that the groups' coefficients are
 683 either all active or all equal to zero. Therefore, it will be enough for one channel
 684 to exhibit one peak (real or not) in order to activate a full group, and therefore
 685 to select an area by consensus. Again, this issue can be partly addressed by ad-
 686 ditional post-processing. However, this comparison shows that, though slightly
 687 more computationally intensive, the proposed approach outperforms LASSO
 688 and ordinary group LASSO for the task at hand.

689 5. Conclusion

690 In this paper, we have presented a new method for pulse localization in intra-
 691 cardiac ECGs based on an overlapping group LASSO followed by a consensus-
 692 based post-processing step. A novel way to implement the proposed solution
 693 efficiently was also derived. Results on synthetic and real data have shown that
 694 the proposed approach is well suited for pulse localization in this framework,
 695 and is more respectful of the physiological constraints inherent to this type of
 696 signals. Further work will focus on developing new penalties related to the peri-
 697 odicity and physiological constraints of the signals, as well as adaptive methods
 698 to take into account the different statistical properties of the individual chan-
 699 nels. Additional experiments and quantitative validation on both intracardiac
 700 and external ECG signals will also be part of future contributions.

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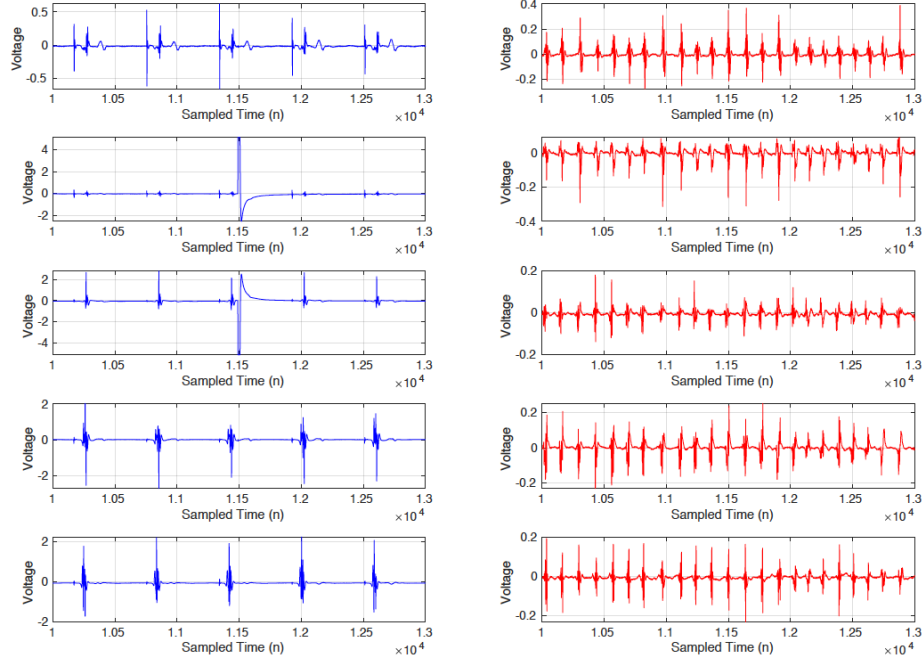


Figure 1: Difference between 5 channels for a normal rhythm (left, blue graphs) and atrial fibrillation rhythm (right, red graphs). Data provided by Dr. Angel Arenal's team, from *Hospital Universitario Gregorio Marañón*, Madrid (Spain).

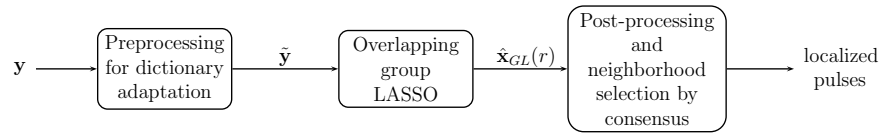


Figure 2: Overall description of the proposed approach.

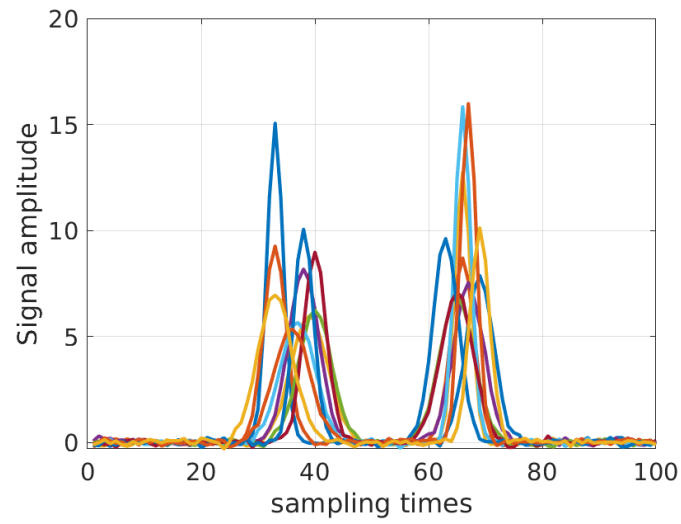


Figure 3: Example of generated multichannel signals (displayed on the same axis for comparison).

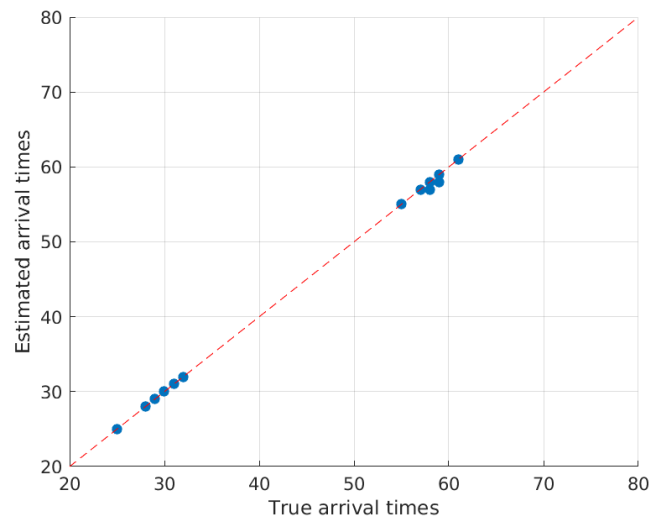


Figure 4: Comparison between estimated times and arrival times for the multichannel signal displayed in Figure 3.

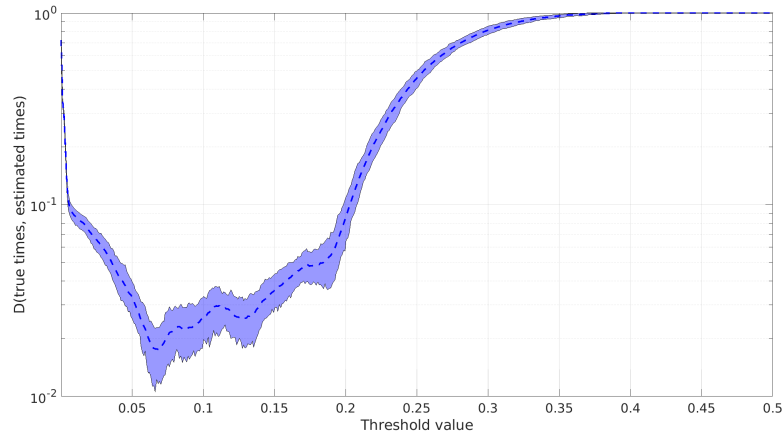


Figure 7: Average D between the true and estimated times with 90% confidence interval for different thresholds - individual strategy.

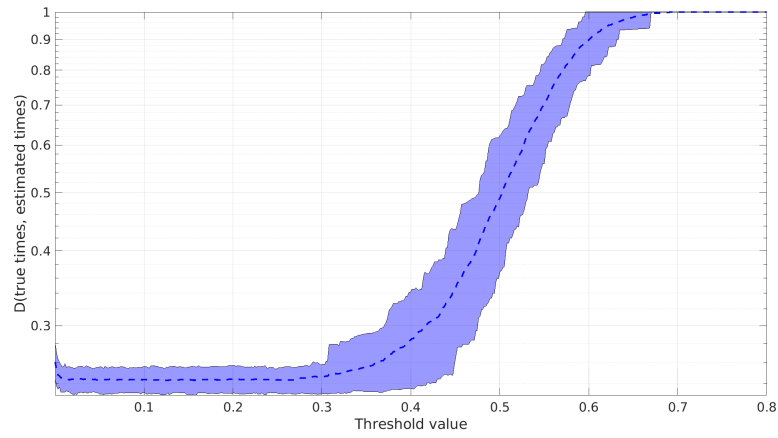


Figure 8: Average D between the true and estimated times with 90% confidence interval for different thresholds - consensus strategy.

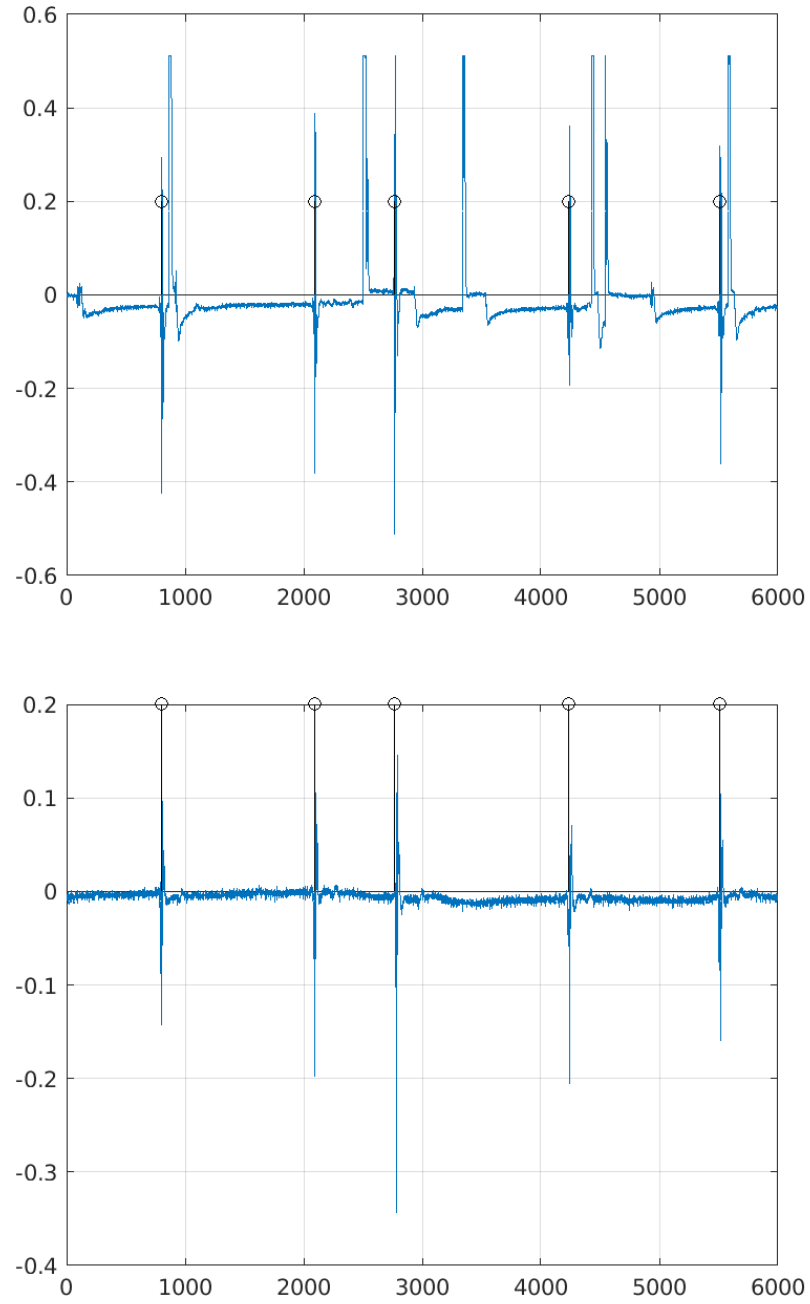


Figure 9: Intracardiac ECG with sinusal rhythm - time estimation by consensus selection.
Results on channels 1 and 2

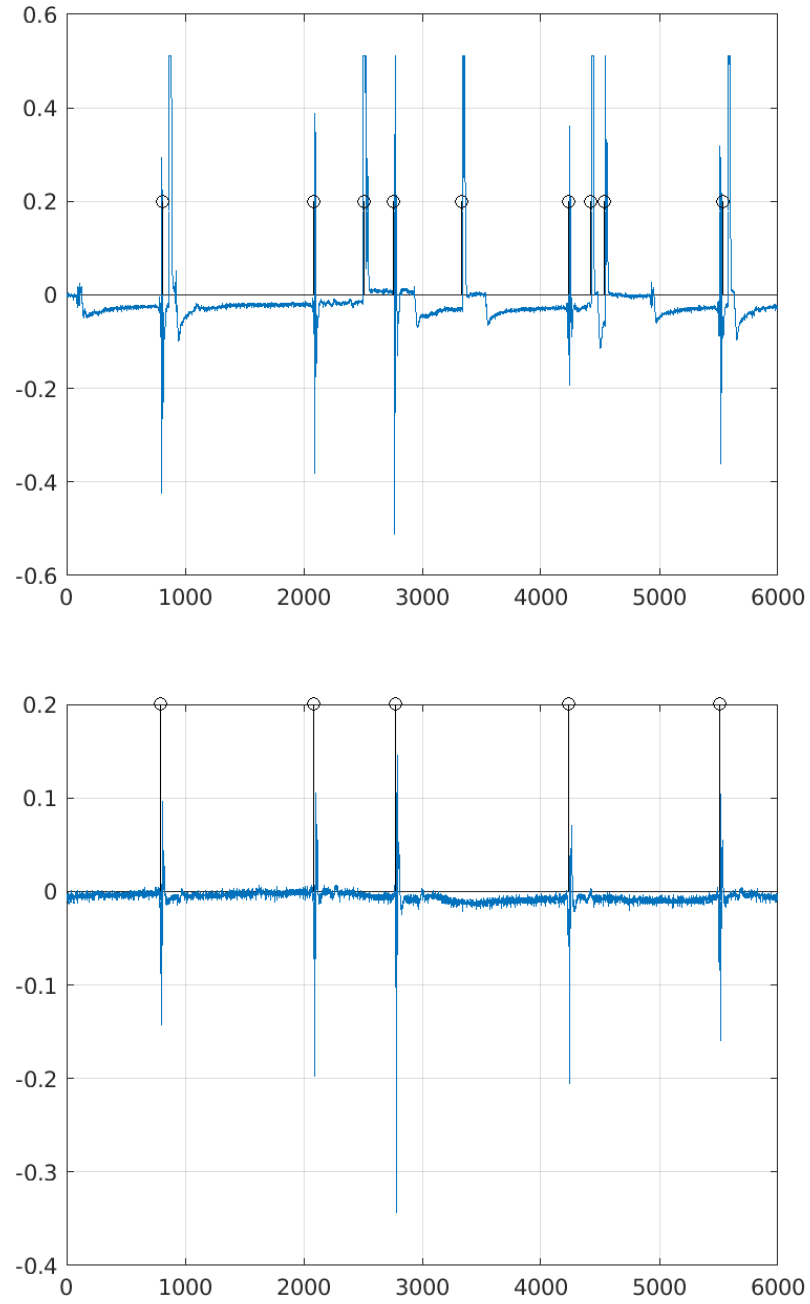


Figure 10: Intracardiac ECG with sinus rhythm - time estimation for each channel individually. Results on channels 1 and 2.

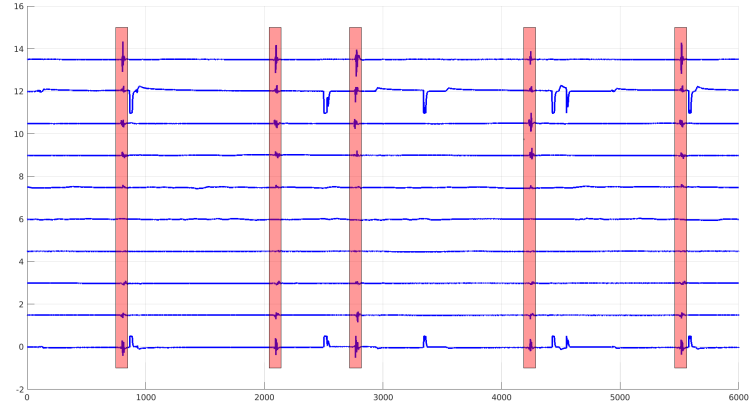


Figure 11: Intracardiac ECG with sinus rhythm - neighborhood estimation by consensus selection.

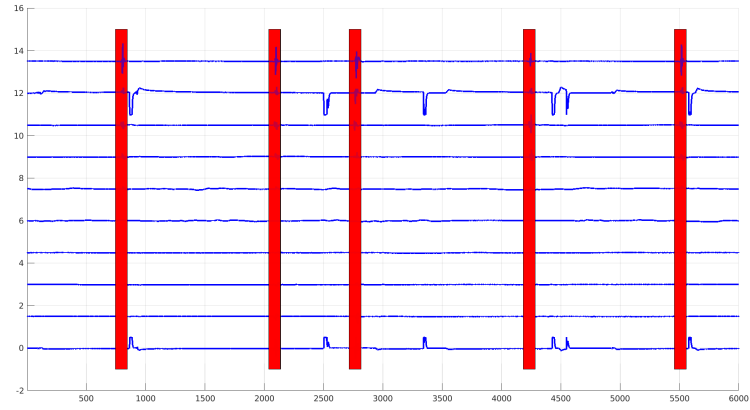


Figure 12: Intracardiac ECG with sinus rhythm - neighborhood estimation for each channel individually - darker shades of red illustrate overlapping intervals.

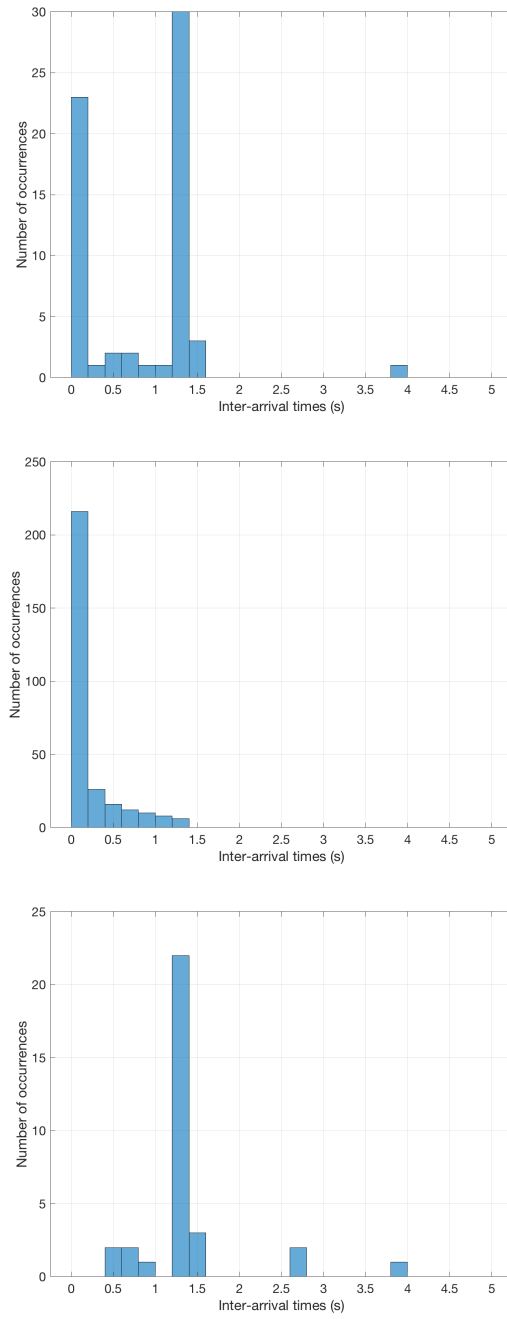


Figure 13: Histograms for sinusal rhythm of the interarrival times using LASSO (up), Group LASSO (middle) and Overlapping Group LASSO (down).

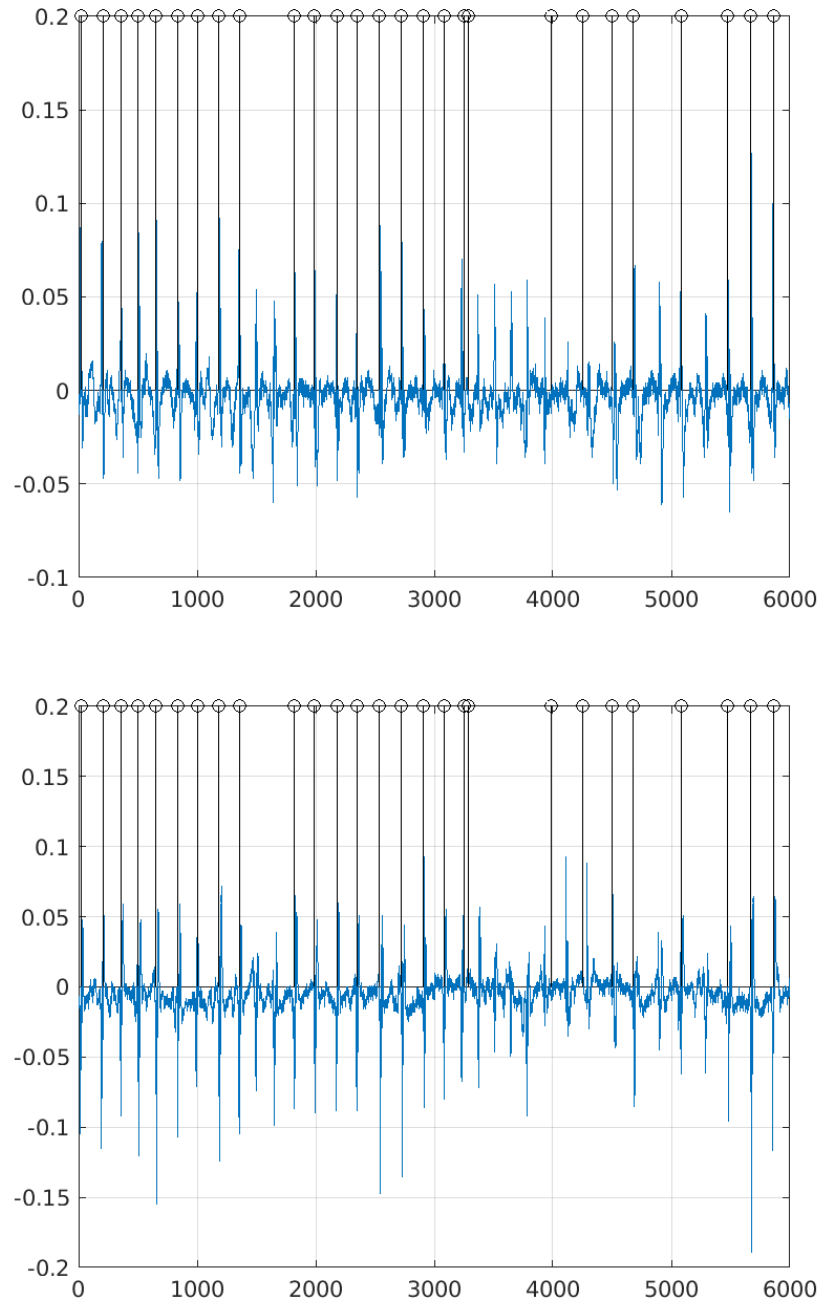


Figure 14: Intracardiac ECG with AF - time estimation by consensus selection. Results on channels 1 and 2.

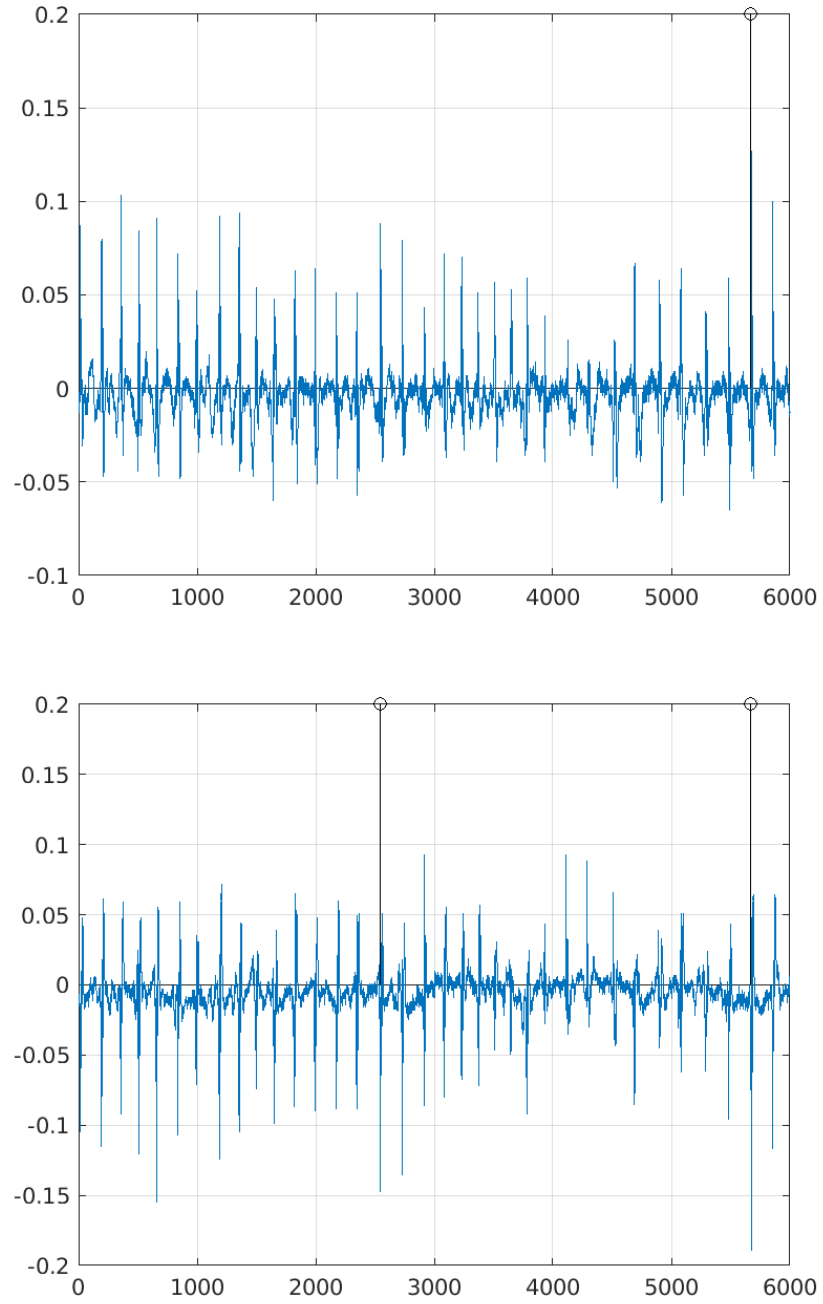


Figure 15: Intracardiac ECG with AF - time estimation for each channel individually. Results on channels 1 and 2.

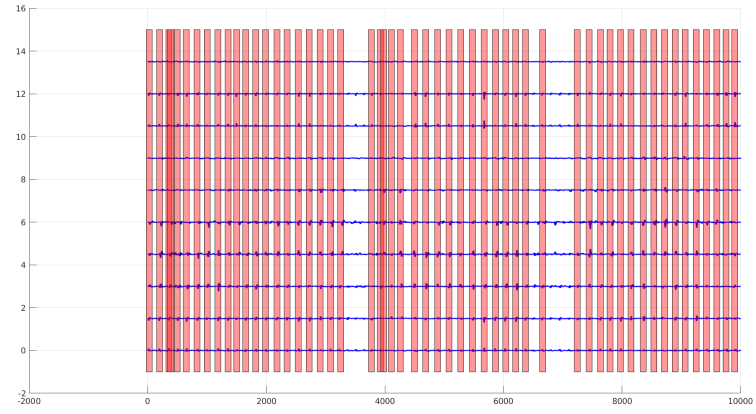


Figure 16: Intracardiac ECG with AF - neighborhood estimation by consensus selection.

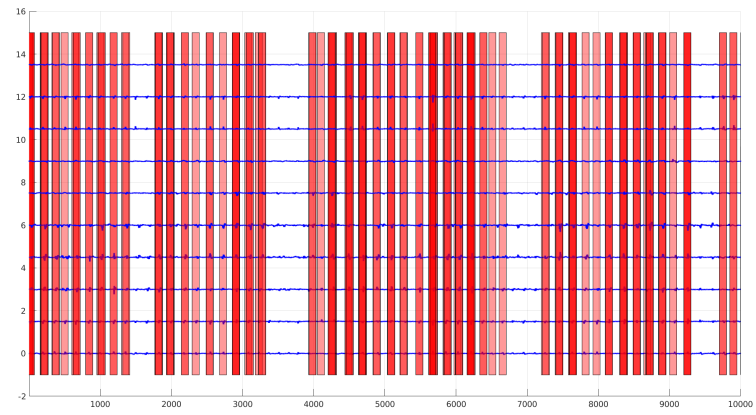


Figure 17: Intracardiac ECG with AF - neighborhood estimation for each channel individually
- darker shades of red illustrate overlapping intervals

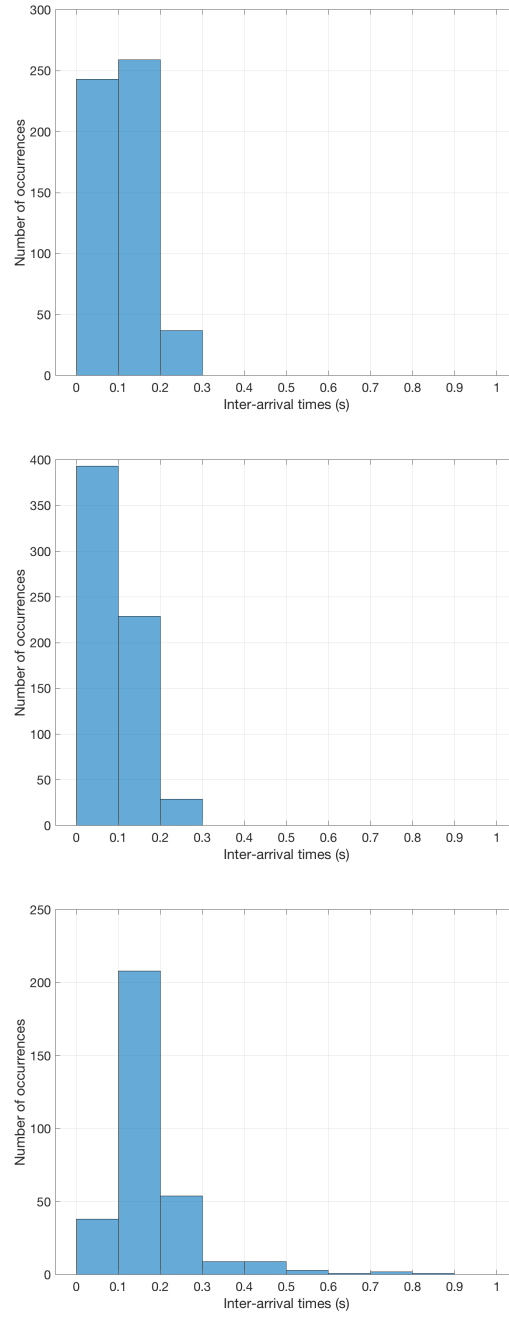


Figure 18: Histograms of the interarrival times for AF using LASSO (up), Group LASSO (middle) and Overlapping Group LASSO (down).

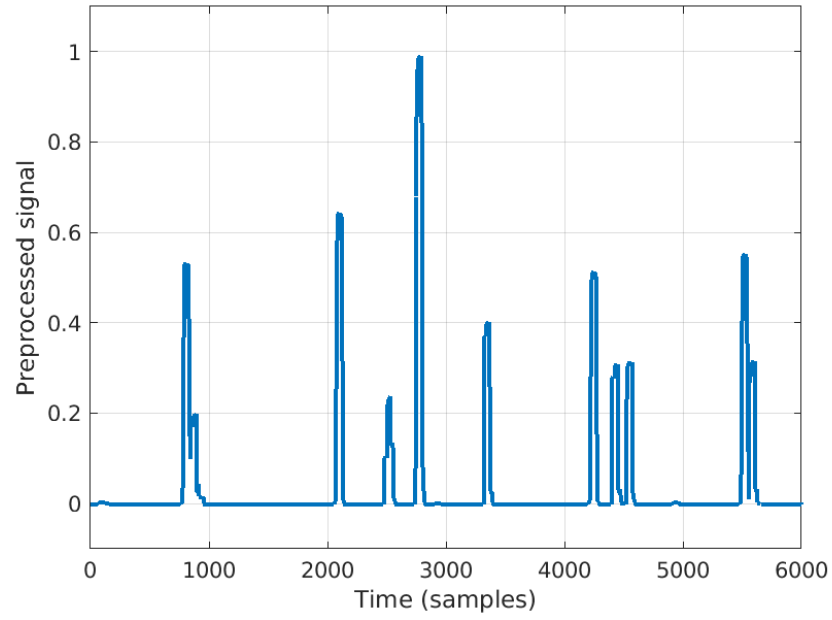


Figure 19: Example of 6000 sample points of one single channel after Pan-Tompkins preprocessing

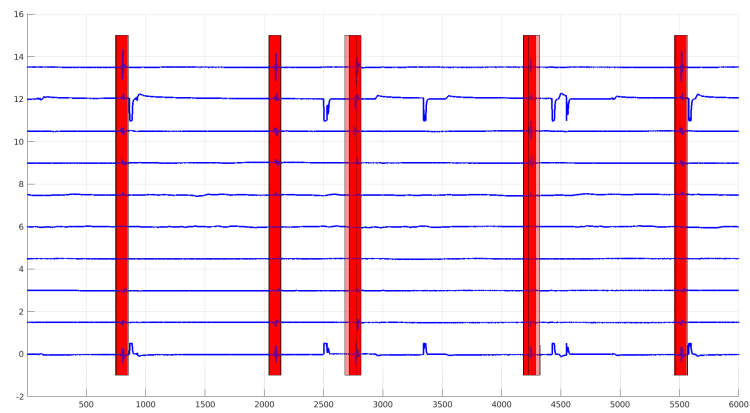


Figure 20: Pulse localization obtained with ordinary LASSO and consensus selection on sinus rhythm data - opaque red areas indicate that several time windows are selected at close times.

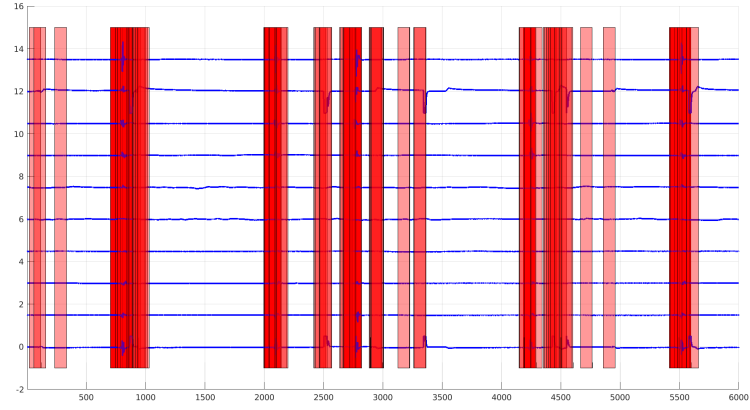
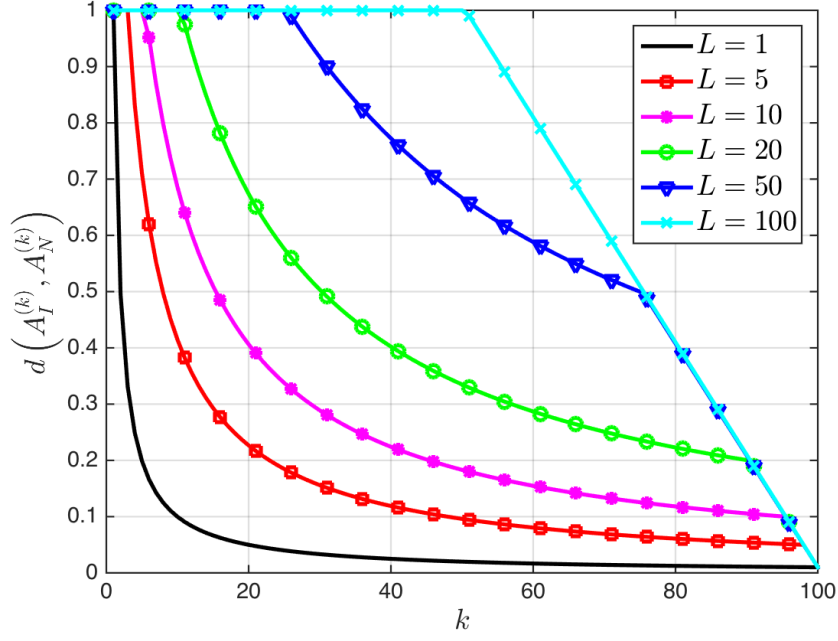
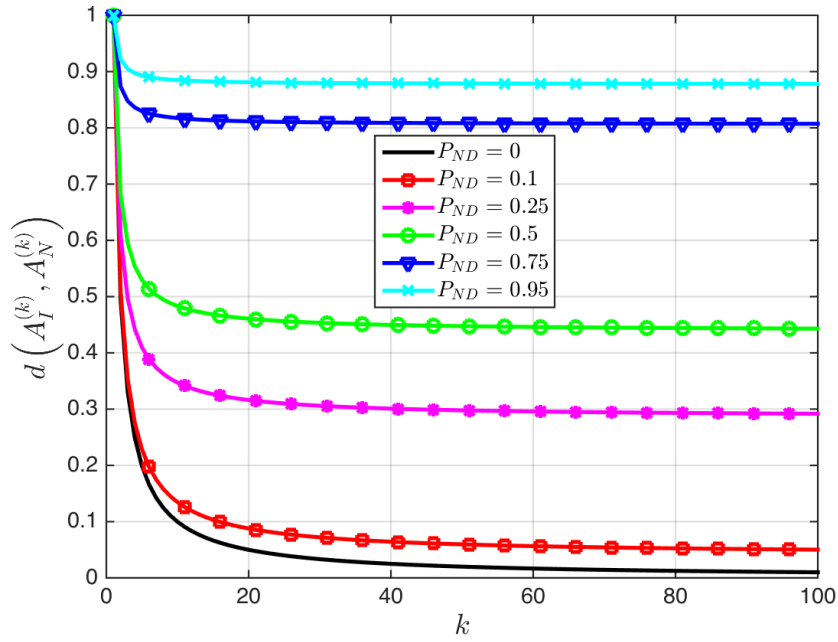


Figure 21: Pulse localization obtained with group LASSO without overlaps and consensus selection on sinus rhythm data - opaque red areas indicate that several time windows are selected at close times.

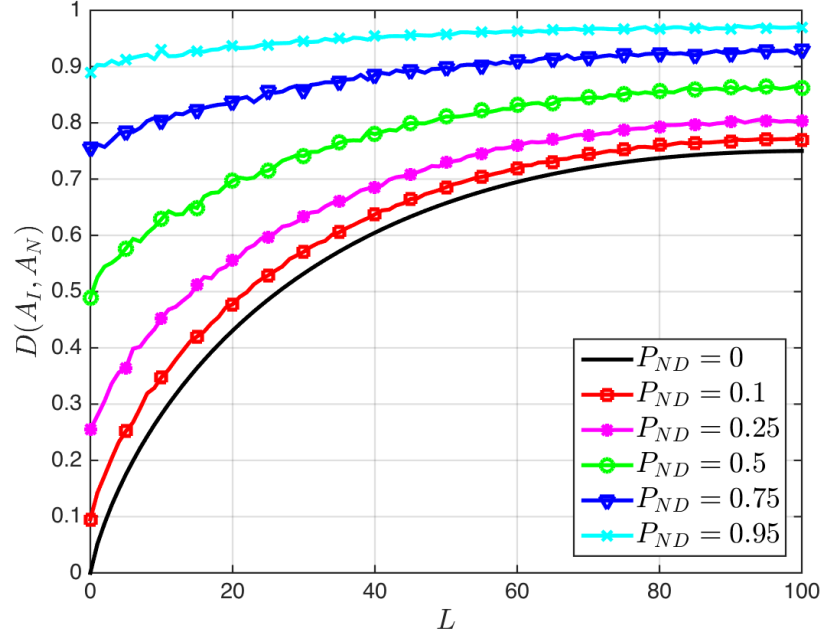


(a)

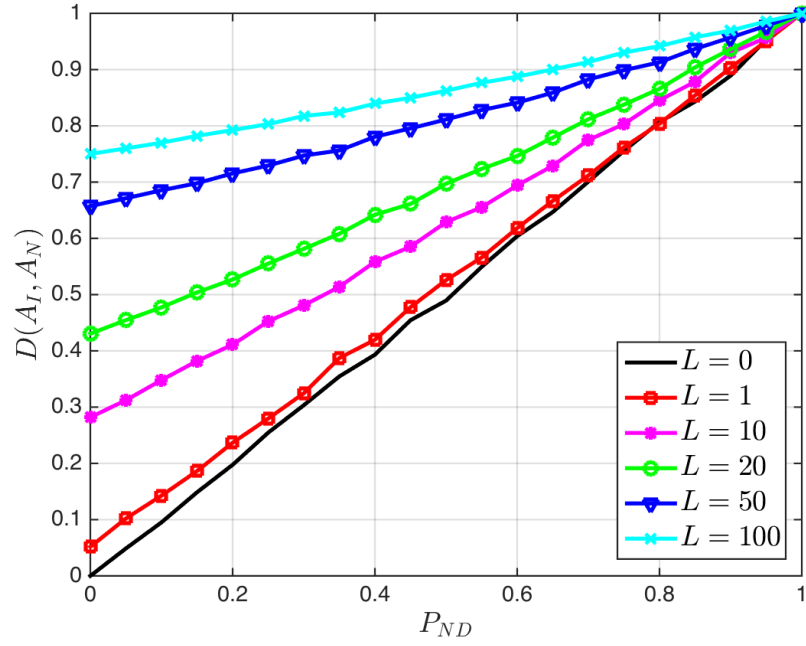


(b)

Figure 5: Evolution of $d(A_I^{(k)}, A_N^{(k)})$ as a function of k . **(a)** For different values of the lag (setting $P_{ND} = 0$). **(b)** For different values of the non-detection probability (setting $L = 1$).



(a)



(b)

Figure 6: $D(A_I, A_N)$. **(a)** As a function of L for different values of P_{ND} . **(b)** As a function of P_{ND} for different values of L .